Fixed-Interior Sharp Minimax Rates for Average Treatment Effects with Unrestricted Discrete Confounding

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Abstract

This paper studies average treatment effect estimation with binary treatment and outcome when the covariate is a discrete variable with d categories and otherwise unrestricted marginal distribution. Under iid sampling and uniform overlap, the estimand is the covariate-adjusted ATE, written as a sum of homogeneous four-cell functionals. For each fixed overlap constant $0 < \epsilon < 1/2$, there are constants $\rho_\epsilon > 0$ and $N_\epsilon < \infty$ such that, for $n \geq N_\epsilon$, $d > 0$, and $d \leq \rho_\epsilon n \log n$, we establish the minimax mean-squared-error rate

$$\mathsf{R}_{n,d,\epsilon} \asymp_\epsilon \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

Under the same fixed-interior calibration conditions, the upper bound is attained by a computable split-sample hybrid estimator: empirical ratios are used on pilot-certified heavy categories, while light categories are estimated by a Chebyshev-polynomial continuation implemented through factorial moments. The lower-bound scale is imported from the fixed-sample construction of Zeng et al. [2024]; the matching computable upper bound, the fixed-interior rate statement, and the resulting parametric-rate and consistency regimes are proved in this paper. The result implies a parametric-rate window $d = O(\sqrt{n} \log n)$, within the fixed-interior range, and a consistency threshold $d = o(n \log n)$. For each fixed interior $0 < \epsilon < 1/2$, under the same hybrid calibration range $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, we also record an oracle-calibrated deterministic comparison envelope near exact randomization. This theoretical comparison uses the known constants ϵ and C_ϵ. At the separate endpoint $\epsilon = 1/2$, the minimax risk is bracketed between constants times n^{-1}, uniformly in d.

1 Introduction

Covariate adjustment is central to observational causal inference. Under consistency, conditional exchangeability, and overlap, the average treatment effect can be identified by comparing treated and control outcomes within covariate strata and averaging over the covariate distribution [Rubin, 1974, Rosenbaum and Rubin, 1983, Hahn, 1998, Hirano et al., 2003]. Classical semiparametric theory describes this problem well when the adjustment dimension is fixed or when nuisance functions can be estimated with enough structure. This paper studies a finite-alphabet benchmark in which that structure is deliberately absent.

The model is simple. A single observation is $O = (X, A, Y)$, where $X \in \{1, \ldots, d\}$, $A \in \{0, 1\}$, and $Y \in \{0, 1\}$. The observations are iid as in assumption 1. The covariate marginal is unrestricted, and treatment assignment satisfies the categorywise overlap condition in assumption 2. With the usual potential-outcome interpretation supplied by assumption 3 and assumption 4, the

ATE is the functional in definition 6. The statistical object is the minimax mean-squared-error risk over the finite-alphabet overlap class in definition 3 and definition 14.

The main result is that, for each fixed $0 < \epsilon < 1/2$, the sharp minimax rate is $n^{-1} + d^2/(n^2 \log^2 n)$ under the stated fixed-interior calibration range. Theorem 1 shows that there are constants depending on ϵ such that, whenever n is large enough, $d > 0$, and $d \le \rho_\epsilon n \log n$,

$$\mathsf{R}_{n,d,\epsilon} \asymp_\epsilon \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

Within the same fixed-interior range, the theorem also gives a parametric-rate window when $d \le K\sqrt{n}\log n$, and it gives the consistency threshold $d = o(n \log n)$. These statements are uniform over unrestricted covariate marginals satisfying only the stated overlap restriction.

The upper bound is constructive. Definition 11 gives a split-sample estimator that first uses pilot counts to classify covariate categories as heavy or light. Heavy categories are estimated by empirical ratios on an independent split. Light categories are estimated by a polynomial approximation to the homogeneous cell functional in definition 5, with factorial moments used to estimate the polynomial terms without unstable empirical denominators. Theorem 2 records both the fixed-overlap risk envelope and the real-arithmetic computability bound for this estimator.

The estimator uses polynomial approximation on weakly sampled cells, a device related to large-alphabet nonsmooth functional estimation [Paninski, 2003, Jiao et al., 2015, Wu and Yang, 2016, Han et al., 2020]. Here the cell contribution is the four-cell ATE term

$$p_k(\mu_{1k} - \mu_{0k}).$$

Zeng et al. [2024] provide the fixed-sample lower-bound ingredient used here and analyze standard regression, weighting, and doubly robust estimators for this high-dimensional discrete ATE problem. This paper constructs the ratio-polynomial hybrid estimator and derives the fixed-interior consistency and parametric-rate regimes. The lower-bound transfer itself is stated in lemma 4.

The result provides a complementary finite-alphabet minimax benchmark for semiparametric and doubly robust theory. In settings with smoothness, sparsity, or accurate nuisance regression and propensity estimates, inverse-probability, doubly robust, targeted, double-machine-learning, and higher-order approaches address different regimes [Robins et al., 1994, Bang and Robins, 2005, van der Laan and Rubin, 2006, Belloni et al., 2014, Chernozhukov et al., 2018, Robins et al., 2008, 2009, 2017, Kennedy et al., 2024]. In the growing unrestricted-alphabet regime studied here, overlap is the sole regularity condition and many small categories create the nonregular term $d^2/(n^2 \log^2 n)$.

The endpoint $\epsilon = 1/2$ is different. At exact within-category randomization, theorem 2 brackets the minimax risk between $1/(100n)$ and $1/n$, uniformly in d. For each fixed interior $0 < \epsilon < 1/2$, and only under the same calibration conditions $n \ge N_\epsilon$ and $d \le \rho_\epsilon n \log n$, the same theorem also gives an oracle-calibrated deterministic comparison envelope near randomization,

$$\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2\right\},$$

up to constants. This comparison chooses between two already analyzed estimators using the known fixed-interior constants and supplies an oracle-calibrated upper envelope near the randomized endpoint.

The formal statements and derivations are supported by a Lean 4 development, with the precise verification boundary recorded in the verification appendix.

2 Setup and assumptions

We work with a finite-alphabet observational model. A single observation is $O_i = (X, A, Y)$, where $X \in \{1, \ldots, d\}$ is a discrete covariate, $A \in \{0, 1\}$ is treatment, and $Y \in \{0, 1\}$ is the observed outcome. The finite-alphabet i.i.d. observational world is deliberately nonparametric in the distribution of X: the cell probabilities may be arbitrary subject only to normalization and the overlap restriction introduced below.

Definition 1. [`def:finite-alphabet-probability-laws`] (Finite-alphabet probability laws Prob) *We say $P \in \mathsf{Prob}(\{1, \ldots, d\} \times \{0, 1\}^2)$ when P is a probability measure on the finite observation alphabet $\{1, \ldots, d\} \times \{0, 1\} \times \{0, \ldots, 1\}$, equivalently a nonnegative mass assignment to each cell (k, a, y) whose total mass over all $k \in \{1, \ldots, d\}$, $a \in \{0, 1\}$, and $y \in \{0, 1\}$ is one.*

For P as in definition 1, write q_{aky} for $P(X = k, A = a, Y = y)$, p_k for $P(X = k)$, π_k for $P(A = 1 \mid X = k)$ when $p_k > 0$, and μ_{ak} for $E_P[Y \mid A = a, X = k]$ when the conditioning event has positive probability. We also write q_k for the four-vector $(q_{0k0}, q_{0k1}, q_{1k0}, q_{1k1})$, ordered by treatment and outcome. The sample size n is fixed throughout the experiment.

Assumption 1. [`ass:iid-sampling`] (Iid sampling) *The observations $O_1, \ldots, O_n$ are independent and identically distributed with single-observation law P.*

assumption 1 is the standard independent and identically distributed sampling condition; it isolates the difficulty of unrestricted discrete confounding from dependence or sampling-design complications [Zeng et al., 2024].

Treatment assignment is summarized category by category.

Definition 2. [`def:bernoulli-law`] (Bernoulli law $\mathsf{Bernoulli}(\pi_k)$) *We say $A \mid X = k \sim \mathsf{Bernoulli}(\pi_k)$ when $A \in \{0, 1\}$ and, conditional on $X = k$, $P(A = 1 \mid X = k) = \pi_k$ and $P(A = 0 \mid X = k) = 1 - \pi_k$.*

Assumption 2. [`ass:overlap`] (Uniform overlap) *For every k with $p_k > 0$,*

$$\epsilon \leq \pi_k \leq 1 - \epsilon.$$

assumption 2 is the standard strong overlap condition, requiring every observed covariate category to retain treatment and control observations with probability bounded away from zero [Zeng et al., 2024]. The constant ϵ may affect constants in the fixed-interior theory, while the estimator defined below uses the same formula across overlap classes.

For causal interpretation, we use the usual potential-outcome overlay only where needed.

Assumption 3. [`ass:consistency`] (Consistency) *The observed outcome satisfies*

$$Y = Y(A)$$

almost surely.

assumption 3 is the standard consistency condition, linking the realized observed outcome to the potential outcome under the assigned treatment [Zeng et al., 2024]. The potential outcomes $Y(a)$ are binary in the present model.

Assumption 4. [`ass:conditional-exchangeability`] (Conditional exchangeability) *The binary potential outcomes satisfy*

$$(Y(0), Y(1)) \perp A \mid X.$$

assumption 4 is the standard conditional exchangeability condition, under which comparisons within a covariate category identify the category-specific causal contrast [Zeng et al., 2024]. Together with assumption 3 and assumption 2, it gives the observational ATE functional its causal interpretation.

The statistical problem is indexed by all finite-alphabet laws obeying these restrictions.

Definition 3. [`def:experiment-class`] (Experiment class $\mathcal{E}_{n,d,\epsilon}$) *Under assumption 1 and assumption 2, the experiment class is*

$$\mathcal{E}_{n,d,\epsilon} := \{P^{\otimes n} : P \in \mathsf{Prob}(\{1,\dots,d\}\times\{0,1\}^2),\ n,d \in \mathbb{N},\ 0<\epsilon\le 1/2,\ P \textit{ satisfies assumption 2}\}.$$

The class $\mathcal{E}_{n,d,\epsilon}$ in definition 3 allows the covariate marginal to vary freely over the d categories. Thus sparsity of categories is part of the estimation problem rather than a regularity condition imposed away.

It is convenient to express each covariate category through its four observed cell masses.

Definition 4. [`def:overlap-cone`] (Overlap cone $\mathcal{C}_\epsilon$) *The overlap-restricted cone of nonnegative four-cell vectors is*

$$\mathcal{C}_\epsilon := \{u=(u_{00},u_{01},u_{10},u_{11}) \in \mathbb{R}^4_+ : \epsilon(u_{00}+u_{01}+u_{10}+u_{11}) \le u_{10}+u_{11} \le (1-\epsilon)(u_{00}+u_{01}+u_{10}+u_{11})\}.$$

definition 4 records overlap as a constraint on a nonnegative four-cell vector $\mathcal{C}_\epsilon$. The cone formulation keeps the category mass and the treatment split in the same object, which is useful when categories have small or vanishing probability.

Definition 5. [`def:cell-functional`] (Cell functional $\phi(u)$) *For $u=(u_{00},u_{01},u_{10},u_{11}) \in \mathcal{C}_\epsilon$, with $\mathcal{C}_\epsilon$ as in definition 4, define*

$$\phi(u) = (u_{00}+u_{01}+u_{10}+u_{11})\left\{\frac{u_{11}}{u_{10}+u_{11}} - \frac{u_{01}}{u_{00}+u_{01}}\right\}$$

when $u \neq 0$, and set $\phi(0,0,0,0)=0$.

The homogeneous contribution ϕ in definition 5 is the category mass multiplied by the treated-minus-control conditional mean contrast. Its homogeneity is the source of the nonsmooth behavior at small cell masses.

Definition 6. [`def:ate-functional`] (ATE functional $\tau(P)$) *The average treatment effect functional is*

$$\tau(P) = \sum_{k=1}^d p_k(\mu_{1k}-\mu_{0k}) = \sum_{k=1}^d \phi(q_k).$$

The estimand $\tau(P)$ in definition 6 is therefore both the usual covariate-adjusted ATE and a sum of four-cell contributions. This second representation is the one used by the estimator and by the minimax analysis.

The estimator separates categories using a deterministic half-sample split.

Definition 7. [`def:balanced-sample-split`] (Balanced sample split I_j) *For $j \in \{0,1\}$, the balanced deterministic half-sample index set $I_j \subseteq \{1,\dots,n\}$ is $I_j = \{i \in \{1,\dots,n\} : i \le \lfloor n/2 \rfloor\}$ when $j=0$, and $I_j = \{i \in \{1,\dots,n\} : \lfloor n/2 \rfloor < i \le n\}$ when $j=1$.*

Definition 8. [`def:split-cell-counts`] (Split cell counts $N^{(0)}_{aky}$ and $N^{(1)}_{aky}$) *For $j \in \{0,1\}$, $k \in \{1,\dots,d\}$, and $a,y \in \{0,1\}$, the split cell count $N^{(j)}_{aky}$ is the number of observations in the deterministic split I_j with covariate value k, treatment value a, and outcome value y:*

$$N^{(j)}_{aky} := \sum_{i\in I_j} \mathbf{1}\{X_i = k,\ A_i = a,\ Y_i = y\}.$$

The first split is used only to classify categories as empirically heavy or light, while the second split estimates their contributions. The count notation $N^{(j)}_{aky}$ in definition 8 will also be used to form factorial moments; for a nonnegative integer z and order r, the falling factorial $(z)_r$ is defined next. In the two definitions that follow, the degree M and the order r are free nonnegative integers; the estimator of definition 11 then fixes the degree at the calibrated value $M = M(n)$ for every sample size $n \geq N_0$, where N_0 is the numerical cutoff recorded there.

Definition 9. [`def:chebyshev-polynomial`] (Chebyshev Polynomial T_M) *We write T_M for the degree-M Chebyshev polynomial of the first kind on $\mathbb{R}$, characterized by $T_0(x) = 1$, $T_1(x) = x$, and $T_{M+2}(x) = 2xT_{M+1}(x) - T_M(x)$ for all $M \geq 0$ and $x \in \mathbb{R}$.*

Definition 10. [`def:falling-factorial`] (Falling factorial $(x)_r$) *For $x, r \in \mathbb{N}_0$, we write $(x)_r$ for the falling factorial of x of order r, defined by $(x)_r = \prod_{\ell=0}^{r-1}(x-\ell)$, with the empty product convention $(x)_0 = 1$; hence $(x)_r = 0$ when $r > x$.*

The Chebyshev polynomial in definition 9 supplies the light-cell approximation, and definition 10 supplies the unbiased factorial-moment representation for its polynomial terms. For the multi-index notation below, $r = (r_{ay})_{a,y\in\{0,1\}}$ ranges over four nonnegative integer coordinates. Throughout, we write $\widehat{\tau}$ for a generic measurable estimator of $\tau(P)$ formed from the sample $O_1, \dots, O_n$; the particular estimator constructed next is $\widehat{\tau}^{\mathrm{hyb}}_n$.

Definition 11. [`def:hybrid-estimator-handle`] (Hybrid estimator $\widehat{\tau}^{\mathrm{hyb}}_n$) *For a sample $O_1, \dots, O_n$ with $O_i = (X_i, A_i, Y_i) \in \{1,\dots,d\} \times \{0,1\} \times \{0,1\}$, set*

$$I_0 = \{1,\dots,\lfloor n/2\rfloor\}, \qquad I_1 = \{\lfloor n/2\rfloor + 1, \dots, n\}, \qquad m_j = |I_j|, \qquad L = \log(en).$$

Let $A = 6$, and set

$$\lambda_0 = 256, \qquad b_0 = 4096, \qquad D_A = 8\log(9A/8) = 8\log(27/4),$$

and

$$\alpha_0 = \min\left\{1, \frac{1}{32\log A}, \frac{1}{256 D_A}\right\}.$$

For $n \geq 2$, define

$$M(n) = \max\{2, \lfloor \alpha_0 L\rfloor\}, \qquad B(n) = b_0 L/m_1.$$

Let N_0 be the least numerical cutoff such that, for every $n \geq N_0$,

$$\alpha_0 L \geq 2, \qquad 4M(n)^2 \leq m_1, \qquad 4M(n)/m_1 \leq 3B(n)/4.$$

For $n \geq N_0$, write $M = M(n)$ and $B = B(n)$, and define the pilot-certified heavy and light category sets by

$$\widehat{\mathcal{H}}_n = \left\{k : \sum_{a,y} N^{(0)}_{aky} > \lambda_0 L\right\}, \qquad \widehat{\mathcal{L}}_n = \widehat{\mathcal{H}}^c_n.$$

For the second split, put

$$N_k^{(1)} = \sum_{a,y} N_{aky}^{(1)}, \qquad N_{ak}^{(1)} = \sum_{y} N_{aky}^{(1)},$$

and define the heavy-category ratio contribution

$$\widehat{h}_k = \frac{N_k^{(1)}}{m_1} \left\{ \frac{N_{1k1}^{(1)}}{N_{1k}^{(1)}} \mathbf{1}\{N_{1k}^{(1)} > 0\} - \frac{N_{0k1}^{(1)}}{N_{0k}^{(1)}} \mathbf{1}\{N_{0k}^{(1)} > 0\} \right\}.$$

Let

$$H_M(x) = \frac{1 - T_M(1-2x)}{2M^2}, \qquad E_M(x) = \frac{H_M(x)}{x}, \qquad G_M(x) = \frac{1 - E_M(x)}{x},$$

where E_M and G_M are interpreted by polynomial continuation at zero. For $u = (u_{00}, u_{01}, u_{10}, u_{11})$, set

$$s_a(u) = u_{a0} + u_{a1}, \qquad s(u) = s_0(u) + s_1(u), \qquad t_1(u) = u_{11}, \qquad t_0(u) = u_{01}.$$

Define the light-cell polynomial approximation

$$P_{M,B}(u) = \frac{s(u)t_1(u)}{B} G_M\left(\frac{s_1(u)}{B}\right) - \frac{s(u)t_0(u)}{B} G_M\left(\frac{s_0(u)}{B}\right).$$

Expanding $P_{M,B}(u) = \sum_{|r| \le M} c_r u^r$, set

$$\widehat{P}_k = \sum_{|r| \le M} c_r \frac{\prod_{a,y} (N_{aky}^{(1)})_{r_{ay}}}{(m_1)_{|r|}}, \qquad \widehat{T}_{\mathcal{H}} = \sum_{k \in \widehat{\mathcal{H}}_n} \widehat{h}_k, \qquad \widehat{T}_{\mathcal{L}} = \sum_{k \in \widehat{\mathcal{L}}_n} \widehat{P}_k.$$

The truncated balanced ratio-polynomial hybrid estimator is

$$\widehat{\tau}_n^{\text{hyb}} = \max\{-1, \min(1, \widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}})\}.$$

For $n < N_0$, set $\widehat{\mathcal{L}}_n = \varnothing$ and use the ratio branch on every category. The definition uses no overlap parameter.

definition 11 combines a ratio estimator on pilot-certified heavy categories with a polynomial estimator on pilot-certified light categories. The sets $\widehat{\mathcal{H}}_n, \widehat{\mathcal{L}}_n$ are determined from I_0, while the light contribution $\widehat{T}_{\mathcal{L}}$ and the final estimator $\widehat{\tau}_n^{\text{hyb}}$ use the second split for estimation. The truncation simply enforces the natural range of an ATE with binary outcomes.

Throughout the reader-facing statistical statements, $\mathbb{N} = \{1, 2, \ldots\}$, so both the sample size and alphabet size are positive. For the operation-count notation set $M(1) = 2$. When $1 \le n < N_0$, the ratio-only branch is evaluated directly; the light-cell quantities $B(n)$, polynomial coefficients, and factorial-moment terms begin at the large-sample branch. The statistical experiment stated here therefore uses positive sample and alphabet sizes, alongside Lean's total conventions at zero.

For comparison near randomized treatment, we also record a centered linear estimator, which we write $\widehat{\tau}_{\text{ctr}}$.

Definition 12. [`def:centered-estimator`] (Centered estimator $\widehat{\tau}_{\mathrm{ctr}}$) *For observations $O_i = (X_i, A_i, Y_i)$, $i = 1, \dots, n$, with binary treatment A_i and binary outcome Y_i, the centered estimator is*

$$\widehat{\tau}_{\mathrm{ctr}} = \frac{2}{n} \sum_{i=1}^{n} (2A_i - 1) \left(Y_i - \frac{1}{2} \right).$$

Definition 13. [`def:selected-estimator`] (Selected estimator $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$) *For fixed $0 < \epsilon < 1/2$ and fixed-interior upper-bound constant C_ϵ, define $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$ to be $\widehat{\tau}^{\mathrm{hyb}}_n$ when*

$$C_\epsilon\{1/n + d^2/(n^2 \log^2 n)\} \leq 1/n + 4(1/2 - \epsilon)^2,$$

and to be $\widehat{\tau}_{\mathrm{ctr}}$ otherwise. At $n = 1$, the quotient involving $\log n$ is interpreted as zero; the asymptotic results use this selector only above their stated large-sample cutoff.

This is an oracle-calibrated deterministic comparison rule. It combines the fixed-interior hybrid bound with the near-randomization linear bound using the fixed overlap constant and the fixed-interior upper-bound constant C_ϵ.

The risk criterion is worst-case mean squared error over the overlap experiment class.

Definition 14. [`def:minimax-risk`] (Minimax risk $\mathsf{R}_{n,d,\epsilon}$) *The minimax mean-squared-error risk is*

$$\mathsf{R}_{n,d,\epsilon} = \inf_{\widehat{\tau}} \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau} - \tau(P))^2\right],$$

where the infimum ranges over all measurable estimators $\widehat{\tau}$ based on $O_1, \dots, O_n$.

The quantity $\mathsf{R}_{n,d,\epsilon}$ in definition 14 is the object characterized in the main results. The supremum is over product laws in definition 3, so the risk treats the covariate distribution and all conditional outcome probabilities as unknown nuisance features.

We close the setup with a two-category construction showing why covariate adjustment is essential even in the smallest nontrivial alphabet.

Definition 15. [`def:two-category-witness`] (Two-category witness law) *For $d = 2$, set*

$$p_1 = p_2 = \frac{1}{2}, \qquad \pi_1 = \epsilon, \qquad \pi_2 = 1 - \epsilon, \qquad \mu_{01} = \mu_{02} = 0, \qquad \mu_{11} = 1, \qquad \mu_{12} = 0.$$

Define a full-data extension by drawing

$$(Y(0), Y(1)) = (0, \mathbf{1}\{X = 1\}),$$

and then drawing $A \mid X = k \sim \mathsf{Bernoulli}(\pi_k)$.

Proposition 1. [`prop:two-category-confounding`] (Two-Category Confounding Witness) *For ϵ, suppose:*

- ***(Real parameter.)*** *$\epsilon \in \mathbb{R}$.*
- ***(Positive contrast.)*** *$0 < \epsilon$.*
- ***(Overlap range.)*** *$\epsilon \leq 1/2$.*

- ***(Witness law.)*** *Q is the full-data law constructed in definition 15, and P is its observed marginal law.*

Then Q satisfies assumption 3 and assumption 4, the observed law P satisfies assumption 2 with overlap constant ϵ, and the ATE functional of definition 6 obeys

$$\tau(P) = \frac{1}{2}, \qquad E_P[Y \mid A = 1] - E_P[Y \mid A = 0] = \epsilon.$$

proposition 1 separates the causal estimand from the unadjusted observed contrast. The construction satisfies the usual causal restrictions, but the treated and control groups place different weights on the two covariate categories. As a result, the naive contrast can be as small as the overlap constant even though the ATE equals one half.

3 Main results

The setup in definition 14 reduces the statistical question to a worst-case mean-squared-error problem over finite-alphabet overlap classes. The main theorem identifies the rate in the fixed-interior case $0 < \epsilon < 1/2$, for large enough n and within the calibrated growth range $d \leq \rho_\epsilon n \log n$. For each fixed $0 < \epsilon < 1/2$ under the calibrated range, the nonparametric term is $d^2/(n^2 \log^2 n)$, reflecting the cost of binary outcomes and unrestricted covariate heterogeneity in this finite-alphabet benchmark. The lower-bound scale is imported from the fixed-sample construction of Zeng et al. [2024]; this paper proves the matching computable hybrid upper bound.

Theorem 1. [`thm:sharp-minimax-fixed-interior`] (Fixed-Interior Minimax Rate)[1] *Fix $0 < \epsilon < 1/2$. Then there exist constants $a_\epsilon, \rho_\epsilon, C_\epsilon > 0$ and an integer $N_\epsilon < \infty$ such that the following hold.*

- ***(Sharp sandwich.)*** *For every n, d, every single-observation law P, and every sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ as in definition 3, if $d > 0$, $n \geq N_\epsilon$, and*

  $$d \leq \rho_\epsilon n \log n,$$

  then, with $\mathsf{R}_{n,d,\epsilon}$ as in definition 14 and $\widehat{\tau}_n^{\mathrm{hyb}}$ as in definition 11,

  $$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\} \leq \mathsf{R}_{n,d,\epsilon} \leq \sup_{Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_Q\left[\left(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(Q) \right)^2 \right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

- ***(Parametric window.)*** *For every $K > 0$, there exists $C_K > 0$ such that, for every n, d, every single-observation law P, and every sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, if $d > 0$, $n \geq N_\epsilon$,*

  $$d \leq K\sqrt{n} \log n \qquad \text{and} \qquad d \leq \rho_\epsilon n \log n,$$

  then

  $$\frac{a_\epsilon}{n} \leq \mathsf{R}_{n,d,\epsilon} \leq \frac{C_K}{n}.$$

- ***(Consistency threshold.)*** *For any sequences n_j, d_j with $n_j \to \infty$, eventually $d_j > 0$, and eventually*
$$d_j \le \rho_\epsilon n_j \log n_j,$$
one has
$$\mathsf{R}_{n_j,d_j,\epsilon} \to 0 \quad \Longleftrightarrow \quad \frac{d_j}{n_j \log n_j} \to 0.$$

theorem 1 gives three consequences under the displayed fixed-interior range. First, throughout $d \le \rho_\epsilon n \log n$, the minimax risk is of exact order

$$n^{-1} + \frac{d^2}{n^2 \log^2 n}.$$

Second, the parametric rate remains attainable when $d = O(\sqrt{n} \log n)$. Third, within the same displayed range, consistent estimation is possible exactly when $d = o(n \log n)$. Thus the logarithmic factor changes the consistency boundary as well as the constants. The constants in the theorem may depend on the fixed overlap margin, while the estimator in definition 11 uses the same formula across ϵ.

The next theorem records the hybrid computability statement, the fixed-interior upper envelope, and the separate behavior near the randomized endpoint. The hybrid part uses the same estimator for all fixed-interior overlap classes. For each fixed $0 < \epsilon < 1/2$, under $n \ge N_\epsilon$ and $d \le \rho_\epsilon n \log n$, the selected rule in definition 13 is only an oracle-calibrated deterministic comparison between that fixed-interior guarantee and the centered linear estimator from definition 12. Polynomial approximation and adaptation ideas of this form are standard in nonsmooth functional estimation [Cai and Low, 2011, Lepski et al., 1999, Jiao et al., 2015], but here they are applied to the four-cell ATE contribution in definition 5.

Theorem 2. [`thm:overlap-adaptive-universal-hybrid`] (Universal Hybrid Envelope) *The following conclusions hold for the hybrid estimator $\widehat{\tau}_n^{\mathrm{hyb}}$ of definition 11 and the centered estimator $\widehat{\tau}_{\mathrm{ctr}}$ of definition 12, with minimax risk $\mathsf{R}_{n,d,\epsilon}$ as in definition 14.*

- ***(Computability.)*** *There is a constant $K \in \mathbb{N}$, $K > 0$, such that for every $n, d \in \mathbb{N}$ there is a real-arithmetic program which, on the hybrid count input, returns exactly $\widehat{\tau}_n^{\mathrm{hyb}}$ for every sample $O_1, \ldots, O_n$, and whose arithmetic/comparison operation count is at most $K\, d\, M(n)^4$.*

- ***(Fixed-overlap hybrid calibration and envelope.)*** *For every $0 < \epsilon < 1/2$, there exist constants $C_\epsilon > 0$, $\rho_\epsilon > 0$, and $N_\epsilon \in \mathbb{N}$ such that, for all $n, d \in \mathbb{N}$ with $n > 0$, $d > 0$, $n \ge N_\epsilon$, and*
$$d \le \rho_\epsilon n \log n,$$
the hybrid estimator satisfies
$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(P))^2\right] \le C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

[1]**Formalization scope.** This result uses the published conclusion [Zeng et al., 2024] (Theorem 2; Appendix C.5; Lemma 4; Appendix D.2). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

For the deterministic selected estimator obtained from the rule that chooses $\widehat{\tau}_n^{\mathrm{hyb}}$ when

$$C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \le \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and otherwise chooses $\widehat{\tau}_{\mathrm{ctr}}$, one also has

$$\mathsf{R}_{n,d,\epsilon} \le \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2 \right]$$

and

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2 \right] \le \max\{C_\epsilon, 4\} \left\{ \frac{1}{n} + \min \left[\frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right] \right\}.$$

- ***(Centered estimator.)*** *For every $0 < \epsilon \le 1/2$ and every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$,*

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2 \right] \le \frac{1}{n} + 4(1/2 - \epsilon)^2.$$

- ***(Endpoint bracket.)*** *For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$,*

$$\frac{1}{100n} \le \mathsf{R}_{n,d,1/2} \le \frac{1}{n}.$$

theorem 2 separates three issues that are often conflated. The hybrid estimator is computable from the finite set of split counts, with operation count polynomial in $M(n)$ and linear in the alphabet size. For each fixed $0 < \epsilon < 1/2$, the same hybrid formula attains the fixed-overlap upper rate in theorem 1. Near $\epsilon = 1/2$, however, the centered estimator has the sharper envelope $n^{-1} + 4(1/2 - \epsilon)^2$, and at the endpoint $\epsilon = 1/2$ the minimax risk is bracketed at the parametric order n^{-1}. The deterministic selected rule uses the displayed theoretical constants and is therefore a comparison device, not an adaptive estimator for unknown overlap or unknown calibration constants.

4 Discussion and extensions

The fixed-interior result extends the usual semiparametric picture. For fixed d, or more generally throughout the parametric window in theorem 1, the minimax risk is of order n^{-1}, in line with the classical efficiency theory for the average treatment effect under overlap [Hahn, 1998, Hirano et al., 2003]. Allowing the alphabet to grow freely adds the challenge of estimating many small, poorly sampled conditional contrasts, reflected by the additional term $d^2/(n^2(\log n)^2)$.

This distinction explains the hybrid estimator's improvement over direct plug-in and weighting estimators in the unrestricted discrete model. A plug-in estimator based on empirical conditional means treats each category as if its arm-specific denominators were reliable. In light categories, those denominators are often zero or very small, and trimming or stabilization introduces bias whose worst-case size depends on the number of categories. Inverse-probability weighting faces the same local denominator problem from the other side: bounded population overlap in assumption 2 coexists with unstable empirical treatment-cell counts when category mass is small. The hybrid estimator in definition 11 uses ratios on pilot-certified heavy categories and a polynomial continuation on the light categories. The sharp logarithmic factor in theorem 1 confirms that this distinction is essential to the rate.

Doubly robust and targeted estimators are designed to protect first-order inference against nuisance estimation error, and they are central in modern observational causal inference [Robins et al., 1994, Bang and Robins, 2005, van der Laan and Rubin, 2006]. Their usual guarantees are most naturally expressed through convergence conditions on nuisance regressions or propensities. The warning of Robins and Ritov [1997] that unrestricted high-dimensional adjustment can obstruct uniform model-free inference is therefore especially relevant here. The experiment class in definition 3 imposes no smoothness, sparsity, margin, or ordering condition that would make such nuisance rates available uniformly. The minimax statement isolates that difficulty in a finite-alphabet benchmark: for each fixed $0 < \epsilon < 1/2$ under the calibrated range, binary outcomes and unrestricted covariate heterogeneity yield the term $d^2/(n^2(\log n)^2)$.

Higher-order influence-function methods are also closely related in motivation, since they seek to reduce bias from estimating high-dimensional nuisance components [Robins et al., 2008, 2009, 2017, Liu et al., 2017, Kennedy et al., 2024]. The present construction is complementary: it replaces unstable light-cell ratios directly by a polynomial approximation to the cell functional in definition 5. The rate in theorem 1 is achieved by the hybrid estimator, whose light-cell branch uses a degree-$M(n)$ polynomial approximation and factorial-moment plug-in terms.

This result is related to recent discussions of limited overlap and high-dimensional adjustment [D'Amour et al., 2021, Balakrishnan et al., 2023, Jin and Syrgkanis, 2025]. Theorem 1 fixes $0 < \epsilon < 1/2$ and shows that many covariate categories with small marginal mass remain difficult under bounded overlap. Theorem 2 separately characterizes the exact-randomization endpoint $\epsilon = 1/2$, where the minimax risk is bracketed between constants times n^{-1}, uniformly in d.

4.1 Limitations and future work

The selected-envelope part of theorem 2 connects these two regimes without claiming a full transition theory or an adaptive implementation. For each fixed interior $0 < \epsilon < 1/2$, and under the theorem's hybrid calibration range $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, it compares the fixed-overlap hybrid estimator with the centered estimator using the known theoretical constants and gives an upper bound involving

$$\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2\right\}.$$

This is useful as a theoretical envelope when the assignment mechanism is close to randomized, because the centered estimator improves as ϵ approaches $1/2$. The selector depends on ϵ and C_ϵ, so it should not be read as a data-driven procedure when those quantities are unknown. Moreover, the theorem does not assert that this combined upper envelope is minimax sharp for triangular arrays with $\epsilon = \epsilon_n$. The fixed-interior lower bound applies for each fixed $0 < \epsilon < 1/2$, and the endpoint bracket applies at $\epsilon = 1/2$; together they leave open the possibility that the exact minimax transition depends on the joint rates at which d, n, and $1/2 - \epsilon_n$ vary.

Appendices

A proofs and auxiliary lemmas

This appendix records the auxiliary statements used to assemble the upper and lower risk bounds in theorem 1 and theorem 2. The organizing distinction is the one built into definition 11: pilot-certified heavy categories are handled by empirical ratios, while pilot-certified light categories are handled by polynomial continuation and factorial moments. The statements below are given as mathematical inputs to the main theorems, with external approximation and minimax ingredients cited where they enter.

The light-cell part is where the logarithmic factor enters the upper bound. Polynomial approximation of nonsmooth distribution functionals has long been used to improve over direct plug-in estimators in large-alphabet problems [Paninski, 2003, Valiant and Valiant, 2011, 2017, Orlitsky et al., 2016, Jiao et al., 2015, Wu and Yang, 2016, Han et al., 2015, 2020, Wu and Yang, 2019]. Here the same principle is applied to the homogeneous four-cell contribution in definition 5. The factorial-moment representation in definition 11 turns the polynomial $P_{M,B}$ into an estimable light-category contribution without dividing by the small empirical treatment counts that make naive ratios unstable.

Lemma 1. [`lem:light-cell-polynomial`] (Light-Cell Polynomial Rate) *Fix an overlap constant ϵ with $0 < \epsilon \le 1/2$. The hybrid estimator is computable in the real-arithmetic model: there is a universal operation-count constant $K > 0$ such that, for every n and d, a real-arithmetic program reading the hybrid count vector returns $\widehat{\tau}_n^{\text{hyb}}$ exactly and uses at most $KdM(n)^4$ arithmetic and comparison operations.*

Moreover, there exist constants $C_\epsilon, c_\epsilon > 0$, depending only on ϵ, such that for every sample size n, alphabet size d, law P on $\{1, \ldots, d\} \times \{0, 1\} \times \{0, 1\}$, and sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ as in definition 3, if

$$d \le c_\epsilon n \log n,$$

then the light-cell contribution satisfies

$$E_{P^{\otimes n}} \left[\left\{ \widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) \right\}^2 \right] \le C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

Proof of lemma 1. Computability. The program reads the $8d$ split counts $N_{aky}^{(j)}$ of definition 8 and evaluates the expression tree obtained by transcribing definition 11 literally; a ratio with vanishing denominator returns 0, which is what the indicators $\mathbf{1}\{N_{ak}^{(1)} > 0\}$ in $\widehat{h}_k$ prescribe. We count one operation for each arithmetic operation and each comparison, and none for reading an input or a constant. Recall $M = M(n) \ge 2$. Building up from the innermost term:

- a falling factorial $(N_{aky}^{(1)})_{r_{ay}}$ costs $4r_{ay}$ operations, four per factor $N_{aky}^{(1)} - \ell$; the four-cell product followed by the division by $(m_1)_{|r|}$ therefore makes one factorial monomial $\prod_{a,y} (N_{aky}^{(1)})_{r_{ay}} / (m_1)_{|r|}$ cost $4|r| + 5$ operations;
- the monomials that occur in the expansion of $P_{M,B}$ are indexed by an arm a, a cell (a', y'), and integers $0 \le t \le j \le M - 2$, and the corresponding multi-index has $|r| = j + 2$, so one coefficient-times-monomial term costs $4j + 14$ operations;

- summing the four cells costs $16j+60 \le 50M$; summing over $t \le j$ costs at most $M(50M+1) \le 51M^2$; summing over $j \le M-2$ costs at most $(M-1)(51M^2+1) \le 52M^3$; hence each arm costs at most $52M^3$, and $\widehat{P}_k = (\text{arm } 1) - (\text{arm } 0)$ costs at most $2 \cdot 52M^3 + 1 \le 105M^4$;

- for one category k, forming the pilot count $\sum_{a,y} N^{(0)}_{aky}$ costs 4, comparing it with $\lambda_0 L$ costs 2, and the ratio contribution $\widehat{h}_k$ costs 11; so a category costs at most $105M^4+17 \le 110M^4$ when $n \ge N_0$, and exactly $11 \le 110M^4$ when $n < N_0$, where every category takes the ratio branch;

- adding the d category contributions costs at most $d \cdot 110M^4 + d \le 111\, dM^4$, and the truncation $\max\{-1, \min(1, \cdot)\}$ multiplies the count by four and adds six, giving at most $4 \cdot 111\, dM^4 + 6 \le 450\, dM^4$ because $dM^4 \ge 1$.

For $d = 0$ the program is the constant 0 and costs nothing. In both cases the program returns $\widehat{\tau}^{\mathrm{hyb}}_n$ exactly, so the first assertion holds with the universal constant $K = 450$.

Risk bound. Write $\mathsf{r}_{n,d} = 1/n + d^2/\{n^2(\log n)^2\}$ for the target rate, put

$$K_c = 4(e+1)8192^2 + 16 \cdot 8192^2, \qquad K_b = \left(\frac{65536}{\epsilon\alpha_0^2}\right)^2, \qquad K_f = 8(e+1)8192^2,$$

and take $C_\epsilon = 3(K_c + K_b + K_f)$ and $c_\epsilon = 1$. These are positive and depend only on ϵ, since α_0 is the numerical constant of definition 11. Fix n, d, P and a sample law in $\mathcal{E}_{n,d,\epsilon}$; by definition 3 that law is $P^{\otimes n}$ and P obeys assumption 2 with constant ϵ. All expectations below are under $P^{\otimes n}$, equivalently under the law of an infinite iid sequence with marginal P, because every integrand depends only on $O_1, \ldots, O_n$; all of them are finite sums over a finite sample space, so every expectation exists and integration is linear and monotone.

If $n < N_0$, then $\widehat{\mathcal{L}}_n = \varnothing$ by definition 11, so $\widehat{T}_{\mathcal{L}} = 0$ and $\sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) = 0$; the left-hand side vanishes and $C_\epsilon \mathsf{r}_{n,d} \ge 0$.

Assume from now on $n \ge N_0$, so that the three calibration inequalities of definition 11 are available:

$$\alpha_0 L \ge 2, \qquad 4M^2 \le m_1, \qquad \frac{4M}{m_1} \le \frac{3B}{4}.$$

Four consequences are used repeatedly. First, $D_A = 8\log(27/4) > 8$ because $\log(27/4) > 1$, hence $\alpha_0 \le 1/(256 D_A) \le 1/120$ and therefore

$$L \;\ge\; \frac{2}{\alpha_0} \;\ge\; 240.$$

Second, $m_1 = n - \lfloor n/2 \rfloor \ge n/2$, so

$$B = \frac{4096\, L}{m_1} \le \frac{8192\, L}{n}, \qquad \text{and} \qquad \frac{1}{m_1} \le \frac{2}{n}.$$

Third, $\log n \le L \le 2\log n$ and $M \le L$, and the calibration also gives the growth inequality

$$6^{4M} L^6 \le n.$$

Fourth, $\alpha_0 L \ge 2$ forces $\lfloor \alpha_0 L \rfloor \ge 2$, so $M = \lfloor \alpha_0 L \rfloor \ge \alpha_0 L - 1 \ge \alpha_0 L/2$, i.e.

$$M \;\ge\; \frac{\alpha_0 L}{2}.$$

The three error terms. Split the categories by their population mass into the deterministic sets

$$\mathcal{G}_n = \{k \in \{1,\dots,d\} : p_k \le B/4\}, \qquad \mathcal{G}_n^c = \{k \in \{1,\dots,d\} : p_k > B/4\},$$

the genuinely light categories and the rest; $\mathcal{G}_n$ depends on P only, whereas the selected set $\widehat{\mathcal{L}}_n$ is random. Every factorial monomial appearing in $\widehat{P}_k$ has multi-index of total order at most $M \le 4M^2 \le m_1$, and for $|r| \le m_1$ the ratio $\prod_{a,y}(N^{(1)}_{aky})_{r_{ay}}/(m_1)_{|r|}$ is an unbiased estimator of the monomial $\prod_{a,y} q_{aky}^{r_{ay}}$; summing the expansion of $P_{M,B}$ term by term therefore gives, for every category k,

$$E_{P^{\otimes n}}\left[\widehat{P}_k\right] = P_{M,B}(q_k).$$

Define

$$\begin{aligned}
X &= \sum_{k\in\mathcal{G}_n} \mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\left\{\widehat{P}_k - P_{M,B}(q_k)\right\},\\
Y &= \sum_{k\in\mathcal{G}_n} \mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\left\{P_{M,B}(q_k) - \phi(q_k)\right\},\\
Z &= \sum_{k\in\mathcal{G}_n^c} \mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\left\{\widehat{P}_k - \phi(q_k)\right\}.
\end{aligned}$$

Since the selection indicator annihilates the categories outside $\widehat{\mathcal{L}}_n$, splitting the sum over all categories at $\mathcal{G}_n$ and inserting $\pm P_{M,B}(q_k)$ on $\mathcal{G}_n$ gives the pointwise identity

$$\widehat{T}_{\mathcal{L}} - \sum_{k\in\widehat{\mathcal{L}}_n}\phi(q_k) = \sum_{k=1}^{d} \mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\left\{\widehat{P}_k - \phi(q_k)\right\} = X + Y + Z.$$

Bound for X. Every $k \in \mathcal{G}_n$ satisfies $p_k + 4M/m_1 \le B/4 + 3B/4 = B$ by the third calibration inequality, and $4M^2 \le m_1$; for such categories the factorial-moment estimator obeys the variance and covariance bounds

$$\begin{aligned}
E_{P^{\otimes n}}\left[\left\{\widehat{P}_k - P_{M,B}(q_k)\right\}^2\right] &\le 8(e+1)\left(B6^M\right)^2,\\
E_{P^{\otimes n}}\left[\left\{\widehat{P}_k - P_{M,B}(q_k)\right\}\left\{\widehat{P}_l - P_{M,B}(q_l)\right\}\right] &\le \frac{8M^2}{m_1}\left(B6^M\right)^2
\end{aligned}$$

for $k \ne l$, the factor 6^M being the coefficient envelope of $P_{M,B}$ on categories of mass at most B. The selection indicators are measurable functions of the pilot split I_0 and the centered errors are functions of the estimation split I_1, so the two factor in each pairwise expectation, and the indicator factor lies in $[0,1]$; expanding X^2 into its $|\mathcal{G}_n|^2$ pairs and using $|\mathcal{G}_n| \le d$ gives

$$E_{P^{\otimes n}}\left[X^2\right] \le |\mathcal{G}_n|\cdot 8(e+1)\left(B6^M\right)^2 + |\mathcal{G}_n|^2\cdot\frac{8M^2}{m_1}\left(B6^M\right)^2 \le 8(e+1)8192^2\,\frac{d\,L^2 6^{2M}}{n^2} + 16\cdot 8192^2\,\frac{d^2 L^4 6^{2M}}{n^3},$$

where the second step used $B \le 8192L/n$, $M \le L$ and $1/m_1 \le 2/n$. The growth inequality $6^{4M}L^6 \le n$ together with $0 < \log n \le L$ converts these into rate terms,

$$\frac{d\,L^2 6^{2M}}{n^2} \le \frac{\mathsf{r}_{n,d}}{2}, \qquad \frac{d^2 L^4 6^{2M}}{n^3} \le \mathsf{r}_{n,d},$$

so that

$$E_{P^{\otimes n}}\left[X^2\right] \le K_c\, \mathsf{r}_{n,d}.$$

Bound for Y. Fix $k \in \mathcal{G}_n$. The four-cell vector q_k has total mass $s(q_k) = p_k \le B/4 \le B$, and it lies in the cone $\mathcal{C}_\epsilon$ of definition 4: on categories with $p_k > 0$ this is assumption 2 written as $\epsilon s(q_k) \le s_1(q_k) \le (1-\epsilon)s(q_k)$, and on categories with $p_k = 0$ all four coordinates vanish. The Chebyshev polynomial of definition 9 satisfies $T_M(1-2x) = 1 - 2M^2x + 2M^2x^2G_M(x)$ for the continuation G_M of definition 11, and $|T_M| \le 1$ on $[-1,1]$, whence the reciprocal-approximation certificate

$$x\,|1 - xG_M(x)| \le \frac{1}{M^2}, \qquad 0 \le x \le 1.$$

Applying it at $x = s_a(u)/B \in [0,1]$ for each arm a, and using $t_a(u) \le s_a(u)$ and $s_a(u) \ge \epsilon s(u)$ from definition 4, the two arms of $P_{M,B}$ each approximate the corresponding arm of ϕ to within $B/(\epsilon M^2)$, so

$$|P_{M,B}(u) - \phi(u)| \le \frac{2B}{\epsilon M^2} \qquad \text{for all } u \in \mathcal{C}_\epsilon \text{ with } s(u) \le B.$$

Since the selection indicator is bounded by one and $|\mathcal{G}_n| \le d$, the triangle inequality gives, pointwise in the sample,

$$|Y| \le \sum_{k\in\mathcal{G}_n} |P_{M,B}(q_k) - \phi(q_k)| \le d\,\frac{2B}{\epsilon M^2}.$$

Now insert the calibration bounds $2B \le 16384L/n$ and $M \ge \alpha_0 L/2$, and then $\log n \le L$:

$$\frac{2B}{\epsilon M^2} \le \frac{16384\, L/n}{\epsilon\,(\alpha_0 L/2)^2} = \frac{65536}{\epsilon\alpha_0^2\, nL} \le \frac{65536}{\epsilon\alpha_0^2\, n\log n}.$$

Squaring $|Y| \le 65536\, d/(\epsilon\alpha_0^2 n\log n)$ gives, pointwise in the sample,

$$Y^2 \le \left(\frac{65536}{\epsilon\alpha_0^2}\right)^2 \frac{d^2}{n^2(\log n)^2} \le K_b\, \mathsf{r}_{n,d},$$

and taking expectations of a pointwise bound by a constant preserves it: $E_{P^{\otimes n}}[Y^2] \le K_b\, \mathsf{r}_{n,d}$.

Bound for Z. Fix k with $p_k > B/4$. Such a category is selected as light only when its pilot count on I_0 falls below $\lambda_0 L$, an event of probability at most $\exp\{-\lfloor n/2\rfloor p_k/8\}$ by the Bernoulli lower-tail estimate. The selection indicator is a function of I_0 and the cell error a function of I_1, so they factor, and the unselected second moment is controlled by the coefficient envelope of $P_{M,B}$ at a category of mass p_k:

$$E_{P^{\otimes n}}\left[\mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\left\{\widehat{P}_k - \phi(q_k)\right\}^2\right] = \Pr\left(k \in \widehat{\mathcal{L}}_n\right)\, E_{P^{\otimes n}}\left[\left\{\widehat{P}_k - \phi(q_k)\right\}^2\right] \le e^{-\lfloor n/2\rfloor p_k/8}\cdot 8(e+1)B^2\left\{6^M\left(1 + \frac{p_k}{B}\right)\right.$$

The pilot tail now absorbs the coefficient growth, uniformly over the whole range $p_k > B/4$. Write $z = p_k/B$, so that $z > 1/4$ and hence $1 + z \le 5z$ and $6(1+z) \le 30z$. Since $\log x \le x - 1$ for $x > 0$,

$$\log\{6(1+z)\} \le 6(1+z) - 1 \le 30z,$$

and this logarithm is nonnegative because $6(1+z) \ge 1$; combining with $M \le L$ gives the growth exponent bound

$$2M \log\left\{6\left(1 + \frac{p_k}{B}\right)\right\} \le 60Lz.$$

For the tail exponent, $p_k = Bz$ with $B = 4096L/m_1$, and $m_1 = n - \lfloor n/2 \rfloor \le 2\lfloor n/2 \rfloor$, so

$$\frac{\lfloor n/2 \rfloor\, p_k}{8} = 512Lz\, \frac{\lfloor n/2 \rfloor}{m_1} \ge 256Lz.$$

Subtracting the two displays and using $z \ge 1/4$ and $L > 0$,

$$-\frac{\lfloor n/2 \rfloor\, p_k}{8} + 2M \log\left\{6\left(1 + \frac{p_k}{B}\right)\right\} \le -196Lz \le -49L,$$

and exponentiating gives the calibration inequality

$$e^{-\lfloor n/2 \rfloor p_k/8} \left\{6^M \left(1 + \frac{p_k}{B}\right)^M\right\}^2 \le e^{-49L}.$$

Multiplying it by $8(e+1)B^2 \ge 0$ therefore yields

$$E_{P^{\otimes n}}\left[\mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\left\{\widehat{P}_k - \phi(q_k)\right\}^2\right] \le 8(e+1)B^2 e^{-49L}.$$

By Cauchy–Schwarz, $Z^2 \le |\mathcal{G}_n^c| \sum_{k \in \mathcal{G}_n^c} \mathbf{1}\{k \in \widehat{\mathcal{L}}_n\}\{\widehat{P}_k - \phi(q_k)\}^2$ pointwise, and $|\mathcal{G}_n^c| \le d$, so

$$E_{P^{\otimes n}}\left[Z^2\right] \le d^2 \cdot 8(e+1)B^2 e^{-49L}.$$

Since $L \ge 1$ and $e^L = en$, we have $e^{-49L} \le e^{-L} \le 1/n$; with $B \le 8192L/n$ this yields

$$E_{P^{\otimes n}}\left[Z^2\right] \le K_f\, \frac{d^2 L^2}{n^3} \le K_f\, \mathsf{r}_{n,d},$$

the last step because $L^2(\log n)^2 \le L^4 \le 6^{4M} L^6 \le n$.

Assembly. From $(X-Y)^2 + (X-Z)^2 + (Y-Z)^2 \ge 0$ one gets the pointwise inequality

$$(X+Y+Z)^2 \le 3\left(X^2 + Y^2 + Z^2\right),$$

so, taking expectations and adding the three bounds,

$$E_{P^{\otimes n}}\left[\left\{\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)\right\}^2\right] \le 3\left(K_c + K_b + K_f\right) \mathsf{r}_{n,d} = C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\},$$

which is the asserted bound. □

lemma 1 has two roles. Its computability clause gives an explicit finite algorithm whose operation count is polynomial in the approximation degree after the split counts are formed. Its risk clause controls the stochastic error and approximation bias of the light side at the target scale. The degree $M(n) \asymp \log n$ is what produces the denominator $(\log n)^2$ in the squared nonparametric term.

The pilot split must classify categories well enough that the two estimators are used on their intended mass ranges. The next statement gives the required sandwich: pilot-heavy categories have population mass above a lower threshold, while pilot-light categories have population mass below an upper threshold, except on an event whose probability is polynomially small.

Lemma 2. [lem:pilot-sandwich] (Pilot Sandwich Bound) *Let $L = \log(en)$. For a sample satisfying assumption 1 with single-observation law P, let $\widehat{\mathcal{H}}$ be the pilot heavy set at threshold t, let $\widehat{\mathcal{L}} = \widehat{\mathcal{H}}^c$, and write $m_0 = |I_0|$, $B_- = tL/(2m_0)$, and $B_+ = 2tL/m_0$.*

For every $K > 0$ and $c > 0$, there exist constants $t_0 > 0$, $C > 0$, and an integer cutoff N such that the following holds. If

- ***(Threshold.)*** *$t \geq t_0$;*
- ***(Sample size.)*** *$n \geq N$;*
- ***(Dimension.)*** *d is a nonnegative integer;*
- ***(Law.)*** *P is a law on the finite observation alphabet $\{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$;*
- ***(Growth.)*** *$d \leq c\, n\, L$;*
- ***(Sampling.)*** *the sample law satisfies assumption 1 with single-observation law P,*

then

$$\Pr\Big(\widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq B_-\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq B_+\}\Big) \leq C n^{-K}.$$

Moreover, for every $c > 0$, there exist $C > 0$ and an integer cutoff N such that the following holds with $t = 256$. If

- ***(Sample size.)*** *$n \geq N$;*
- ***(Dimension.)*** *d is a nonnegative integer;*
- ***(Law.)*** *P is a law on the finite observation alphabet $\{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$;*
- ***(Growth.)*** *$d \leq c\, n\, L$;*
- ***(Sampling.)*** *the sample law satisfies assumption 1 with single-observation law P,*

then

$$\Pr\Big(\widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq 256L/(2m_0)\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq 512L/m_0\}\Big) \leq C n^{-4}.$$

Proof of lemma 2. Throughout, $L = \log(en)$, the first split I_0 of definition 7 has size $m_0 = |I_0| = \lfloor n/2 \rfloor$, and for a category $k \in \{1, \dots, d\}$ we write, with $N^{(0)}_{aky}$ as in definition 8,

$$N^{(0)}_k = \sum_{a,y} N^{(0)}_{aky} = \sum_{i \in I_0} \mathbf{1}\{X_i = k\}, \qquad p_k = P(X = k) = \sum_{a,y} q_{aky},$$

so that $N^{(0)}_k$ is the pilot count of category k and p_k is its population mass. The pilot heavy set at threshold t and its complement are

$$\widehat{\mathcal{H}} = \left\{k : \lfloor tL \rfloor < N^{(0)}_k\right\}, \qquad \widehat{\mathcal{L}} = \widehat{\mathcal{H}}^c = \left\{k : N^{(0)}_k \leq \lfloor tL \rfloor\right\}.$$

Because $N^{(0)}_k$ is an integer, $\lfloor tL \rfloor < N^{(0)}_k$ holds exactly when $tL < N^{(0)}_k$, and $N^{(0)}_k \leq \lfloor tL \rfloor$ exactly when $N^{(0)}_k \leq tL$; at $t = \lambda_0 = 256$ this is the set $\widehat{\mathcal{H}}_n$ of definition 11. Write $\mathcal{B}_t$ for the failure event of the statement, that is, the event that

$$\widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq B_-\} \quad \text{or} \quad \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq B_+\}, \qquad B_- = \frac{tL}{2m_0}, \qquad B_+ = \frac{2tL}{m_0}.$$

On $\mathcal{B}_t$ there is a category k for which either

$$p_k < \frac{tL}{2m_0} \quad \text{and} \quad tL < N_k^{(0)},$$

or

$$\frac{2tL}{m_0} < p_k \quad \text{and} \quad N_k^{(0)} \leq tL.$$

Indeed, if the first inclusion fails there is a $k \in \widehat{\mathcal{H}}$ with $p_k < B_-$, and membership in $\widehat{\mathcal{H}}$ means $\lfloor tL \rfloor < N_k^{(0)}$, i.e. $tL < N_k^{(0)}$; if the second inclusion fails there is a $k \in \widehat{\mathcal{L}}$ with $p_k > B_+$, and membership in $\widehat{\mathcal{L}}$ means $N_k^{(0)} \leq \lfloor tL \rfloor$, i.e. $N_k^{(0)} \leq tL$. Consequently

$$\Pr(\mathcal{B}_t) \leq \sum_{k=1}^{d} \Pr\left(p_k < \frac{tL}{2m_0} \text{ and } tL < N_k^{(0)}\right) + \sum_{k=1}^{d} \Pr\left(\frac{2tL}{m_0} < p_k \text{ and } N_k^{(0)} \leq tL\right).$$

Fix k and assume $n \geq 2$, so that $m_0 = \lfloor n/2 \rfloor \geq 1$. Under assumption 1 the sample law is $P^{\otimes n}$, and the coordinates indexed by I_0 are m_0 independent draws from P; the pilot count is therefore a sum

$$N_k^{(0)} = \sum_{i=1}^{m_0} \mathbf{1}\{X_i = k\}$$

of m_0 independent copies of the indicator $\mathbf{1}\{X = k\}$, which takes values in $\{0, 1\}$ and has population mean

$$E_P[\mathbf{1}\{X = k\}] = p_k.$$

For such a count the exponential Markov inequality gives, for every level a and every tilt $s \geq 0$,

$$\Pr\left(N_k^{(0)} > a\right) \leq \exp\left\{-sa + m_0 p_k (e^s - 1)\right\},$$

and, for every tilt $s \leq 0$,

$$\Pr\left(N_k^{(0)} \leq a\right) \leq \exp\left\{-sa + m_0 p_k (e^s - 1)\right\},$$

because a single indicator with mean p_k has moment generating function $1 + p_k(e^s - 1) \leq \exp\{p_k(e^s - 1)\}$ and the m_0 summands are independent. Taking $s = \log 2$ in the first display and $s = -\log 2$ in the second, so that $e^s - 1$ equals 1 respectively $-1/2$,

$$\Pr\left(N_k^{(0)} > a\right) \leq \exp\left\{-a \log 2 + m_0 p_k\right\}, \qquad \Pr\left(N_k^{(0)} \leq a\right) \leq \exp\left\{a \log 2 - \frac{m_0 p_k}{2}\right\}.$$

Hence, if $m_0 p_k < a/2$, then

$$\Pr\left(N_k^{(0)} > a\right) \leq \exp\left\{-a\left(\log 2 - \tfrac{1}{2}\right)\right\},$$

and if $2a < m_0 p_k$, then

$$\Pr\left(N_k^{(0)} \leq a\right) \leq \exp\left\{-\frac{m_0 p_k}{8}\right\},$$

the second because $2a < m_0 p_k$, $m_0 p_k \geq 0$ and $\log 2 < 3/4$ give $a \log 2 \leq \frac{3}{4} \max\{a, 0\} \leq \frac{3}{8} m_0 p_k$.

Apply these two bounds at the level $a = tL$, with $t \geq 0$. If $p_k < tL/(2m_0)$, then multiplying by $m_0 \geq 1$ gives $m_0 p_k < tL/2$, so

$$\Pr\left(p_k < \frac{tL}{2m_0} \text{ and } tL < N_k^{(0)}\right) \leq \Pr\left(N_k^{(0)} > tL\right) \leq \exp\left\{-tL\left(\log 2 - \tfrac{1}{2}\right)\right\}.$$

If instead $p_k > 2tL/m_0$, then $m_0 p_k > 2tL$, so

$$\Pr\left(\frac{2tL}{m_0} < p_k \text{ and } N_k^{(0)} \leq tL\right) \leq \Pr\left(N_k^{(0)} \leq tL\right) \leq \exp\left\{-\frac{m_0 p_k}{8}\right\} \leq \exp\left\{-\frac{tL}{4}\right\},$$

the last step by monotonicity of the exponential and $m_0 p_k > 2tL$. For a category k whose mass violates the stated condition, the corresponding event is empty and its probability is 0, so every one of the $2d$ terms in the previous sum obeys the displayed bound, and for all $n \geq 2$ and $t \geq 0$,

$$\Pr\left(\mathcal{B}_t\right) \leq d\left[\exp\left\{-tL\left(\log 2 - \tfrac{1}{2}\right)\right\} + \exp\left\{-\frac{tL}{4}\right\}\right].$$

We convert this into a polynomial bound. Let $q > 0$, and suppose $n \geq 1$, $d \leq cnL$, $t \geq 0$, and

$$q \leq t\left(\log 2 - \tfrac{1}{2}\right), \qquad q \leq \frac{t}{4}.$$

Since $n \geq 1$, the scale $L = \log(en) = 1 + \log n$ satisfies

$$1 \leq L \leq n,$$

the upper bound because $\log n \leq n - 1$. Multiplying the two threshold inequalities by $L > 0$ gives $tL(\log 2 - \frac{1}{2}) \geq qL$ and $tL/4 \geq qL$, so both exponential factors are at most e^{-qL}, and therefore

$$d\left[\exp\left\{-tL\left(\log 2 - \tfrac{1}{2}\right)\right\} + \exp\left\{-\frac{tL}{4}\right\}\right] \leq cnL \cdot 2e^{-qL} \leq 2cn^2 e^{-qL},$$

using $d \leq cnL$ and then $L \leq n$. Because $L = \log(en)$,

$$e^{-qL} = \exp\{-q\log(en)\} = e^{-q}n^{-q},$$

so that, using $e^{-q} \leq 1$,

$$2cn^2 e^{-qL} = 2ce^{-q}n^{2-q} \leq 2cn^{2-q}.$$

For the first assertion, let $K > 0$ and $c > 0$ be given. Since $\log 2 > 0.693$ we have $\log 2 - \frac{1}{2} > 0$, so we may set

$$t_0 = \max\left\{\frac{K+4}{\log 2 - 1/2}, 4(K+4)\right\}, \qquad C = 2c, \qquad N = 2,$$

and then $t_0 > 0$ and $C > 0$, because $K + 4 > 0$ and $c > 0$. Let $t \geq t_0$, let $n \geq N = 2$, let $d \leq cnL$, and let the sample law satisfy assumption 1 with single-observation law P, so that it equals $P^{\otimes n}$. Then $t \geq t_0 > 0$ and

$$t\left(\log 2 - \tfrac{1}{2}\right) \geq \frac{K+4}{\log 2 - 1/2}\left(\log 2 - \tfrac{1}{2}\right) = K + 4, \qquad \frac{t}{4} \geq \frac{4(K+4)}{4} = K + 4.$$

So the last two displays apply with $q = K + 4$, and since $2 - q = -K - 2 \le -K$ and $n \ge 1$,

$$\Pr(\mathcal{B}_t) \le 2cn^{-K-2} \le 2cn^{-K} = Cn^{-K}.$$

This is the first assertion.

For the numerical specialization, let $c > 0$ and take $t = 256$, so that the two threshold inequalities hold outright with $q = 4 + 4 = 8$:

$$8 \le 256\left(\log 2 - \tfrac{1}{2}\right), \qquad 8 \le \frac{256}{4} = 64,$$

the first because $\log 2 > 0.693$ gives $256(\log 2 - \frac{1}{2}) > 49$. Hence, with $C = 2c$ and $N = 2$, the same two paragraphs give, for every $n \ge 2$ with $d \le cnL$ and every sample law satisfying assumption 1 with single-observation law P,

$$\Pr(\mathcal{B}_{256}) \le 2cn^{2-8} = 2cn^{-6} \le 2cn^{-4} = Cn^{-4},$$

which is the stated bound at the thresholds $B_- = 256L/(2m_0)$ and $B_+ = 512L/m_0$. □

The constants in lemma 2 are deliberately one-sided. The estimator uses the stated separation to justify ratio bounds on the heavy side and polynomial bounds on the light side. The special threshold $t = 256$ is the calibration used in definition 11.

Once the pilot split supplies that separation, the heavy categories can be treated by ordinary ratio estimation. Overlap controls the arm-specific denominators in population, and the pilot event keeps the selected category masses large enough for the second-split empirical ratios to have the required mean-squared error.

Lemma 3. [lem:universal-heavy-cell-rate] (Heavy Cell Rate) *For every $0 < \epsilon < 1/2$, there exist constants $C_\epsilon, \rho_\epsilon > 0$ and an integer $N_\epsilon < \infty$ such that the following holds. For every $n, d \in \mathbb{N}$, every single-observation law P on $\{1, \dots, d\} \times \{0,1\} \times \{0,1\}$, and every sample law μ_n, assume:*

- ***(Sample size.)*** *$n \ge N_\epsilon$.*
- ***(Dimension growth.)*** *$d \le \rho_\epsilon n \log n$.*
- ***(Experiment class.)*** *$\mu_n = P^{\otimes n}$ belongs to $\mathcal{E}_{n,d,\epsilon}$ in the sense of definition 3.*

Then

$$\int \left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k) \right)^2 d\mu_n \le C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

Proof of lemma 3. Throughout, $L = \log(en)$, $m_0 = |I_0| = \lfloor n/2 \rfloor$ and $m_1 = |I_1| = n - \lfloor n/2 \rfloor$ are as in definition 11, p_k is the mass of category k, $\mu_{ak} = \mathbb{E}_P[Y \mid X = k, A = a]$, q_k is the four-cell vector of category k, and

$$\pi_{ak} = \Pr_P(A = a \mid X = k), \qquad \text{so} \qquad \pi_{1k} = \pi_k, \quad \pi_{0k} = 1 - \pi_k.$$

By assumption 2, $\pi_{ak} \ge \epsilon$ for both arms $a \in \{0,1\}$ whenever $p_k > 0$.

Step 0: the constants. Apply the numerical ($t = 256$, $K = 4$) part of lemma 2 with growth constant $c = 1$. It supplies $C > 0$ and a cutoff N_{bad} such that, writing

$$\mathcal{B} = \left\{ \widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq B_-\} \right\} \cup \left\{ \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq B_+\} \right\}, \qquad B_- = \frac{256L}{2m_0}, \quad B_+ = \frac{512L}{m_0},$$

for the pilot heavy set at threshold 256, one has

$$\Pr_{P^{\otimes n}}(\mathcal{B}) \leq Cn^{-4} \qquad \text{whenever } n \geq N_{\text{bad}} \text{ and } d \leq nL.$$

Set

$$\kappa_\epsilon = \frac{16}{\epsilon} + 12 + \frac{1}{\epsilon^4}, \qquad C_{\text{bad}} = 4C, \qquad C_\epsilon = 4\kappa_\epsilon + C_{\text{bad}},$$
$$\rho_\epsilon = 1, \qquad N_\epsilon = \max\{8, N_0, N_{\text{bad}}\},$$

with N_0 the calibration cutoff of definition 11. Since $\epsilon > 0$ and $C > 0$, both $C_\epsilon > 0$ and $\rho_\epsilon > 0$.

Fix n, d, P, μ_n satisfying the three hypotheses. Membership in $\mathcal{E}_{n,d,\epsilon}$ gives $\mu_n = P^{\otimes n}$ and overlap at ϵ, so all integrals below are against $P^{\otimes n}$. From $n \geq N_\epsilon$ we get $n \geq 8$, $n \geq N_0$ and $n \geq N_{\text{bad}}$; and $\log n \leq L$ turns the dimension hypothesis $d \leq \rho_\epsilon n \log n = n \log n$ into $d \leq nL$, so the pilot bound above is in force. Since $n \geq 8$,

$$m_1 \geq \frac{n}{2} \geq 4, \qquad 2m_0 \leq n, \qquad 4(m_1 - 2) \geq n, \qquad L = 1 + \log n > 0.$$

Abbreviate the heavy error by

$$\mathcal{D} = \widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k),$$

and split the integral over $\mathcal{B}$ and its complement,

$$\int \mathcal{D}^2 \, dP^{\otimes n} = \int_{\mathcal{B}} \mathcal{D}^2 \, dP^{\otimes n} + \int_{\mathcal{B}^c} \mathcal{D}^2 \, dP^{\otimes n}.$$

Step 1: the pilot-bad event. Both $\widehat{T}_{\mathcal{H}}$ and its target lie in $[-1, 1]$. For the estimator, each of the two empirical arm ratios in $\widehat{h}_k$ lies in $[0, 1]$, being a count divided by a larger count (or 0 when the arm count vanishes), so their difference has modulus at most one and

$$|\widehat{h}_k| \leq \frac{N_k^{(1)}}{m_1}, \qquad \text{hence} \qquad |\widehat{T}_{\mathcal{H}}| \leq \sum_{k \in \widehat{\mathcal{H}}_n} \frac{N_k^{(1)}}{m_1} \leq \sum_{k=1}^{d} \frac{N_k^{(1)}}{m_1} = \frac{m_1}{m_1} = 1,$$

the last equality because every index of I_1 is counted in exactly one category.

For the target, evaluating definition 5 at the four cell masses of category k gives

$$\phi(q_k) = p_k(\mu_{1k} - \mu_{0k}), \qquad \text{so} \qquad |\phi(q_k)| \leq p_k,$$

because $\mu_{1k}, \mu_{0k} \in [0, 1]$; summing and using $\sum_{k=1}^{d} p_k = 1$ gives $\left| \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k) \right| \leq 1$.

Consequently $|\mathcal{D}| \leq 2$, so $\mathcal{D}^2 \leq 4$ pointwise and

$$\int_{\mathcal{B}} \mathcal{D}^2 \, dP^{\otimes n} \leq 4 \Pr_{P^{\otimes n}}(\mathcal{B}) \leq 4Cn^{-4} = C_{\text{bad}} n^{-4}.$$

Step 2: the selected categories are heavy in population off $\mathcal{B}$. Because $n \geq N_0$, definition 11 selects $\widehat{\mathcal{H}}_n = \{k : N_k^{(0)} > 256L\}$, and since the pilot counts $N_k^{(0)}$ are integers this is the same set as $\{k : N_k^{(0)} > \lfloor 256L \rfloor\}$, i.e. exactly the pilot heavy set $\widehat{\mathcal{H}}$ at threshold $t = 256$ appearing in $\mathcal{B}$. Hence, off $\mathcal{B}$,

$$p_k \geq B_- = \frac{256L}{2m_0} > 0 \qquad \text{for every } k \in \widehat{\mathcal{H}}_n.$$

Step 3: arm decomposition for a deterministic category set. For a deterministic $H \subseteq \{1, \dots, d\}$ and an arm $a \in \{0, 1\}$, define

$$\widehat{\theta}_a(H) = \sum_{k \in H} \frac{N_k^{(1)}}{m_1} \cdot \frac{N_{ak1}^{(1)}}{N_{ak}^{(1)}} \mathbf{1}\{N_{ak}^{(1)} > 0\}, \qquad \bar{\theta}_a(H) = \sum_{k \in H} \frac{N_k^{(1)}}{m_1} \mu_{ak}, \qquad \theta_a(H) = \sum_{k \in H} p_k \mu_{ak}.$$

By the definition of $\widehat{h}_k$ in definition 11 and by $\phi(q_k) = p_k(\mu_{1k} - \mu_{0k})$ from Step 1,

$$\sum_{k \in H} \widehat{h}_k = \widehat{\theta}_1(H) - \widehat{\theta}_0(H), \qquad \sum_{k \in H} \phi(q_k) = \theta_1(H) - \theta_0(H),$$

so $(x_1 - x_0)^2 \leq 2x_1^2 + 2x_0^2$ applied to $x_a = \widehat{\theta}_a(H) - \theta_a(H)$ gives, pointwise in the sample,

$$\left(\sum_{k \in H} \widehat{h}_k - \sum_{k \in H} \phi(q_k) \right)^2 \leq 2 \left(\widehat{\theta}_1(H) - \theta_1(H) \right)^2 + 2 \left(\widehat{\theta}_0(H) - \theta_0(H) \right)^2.$$

Step 4: the fixed-set arm bound. We claim the following *fixed-set arm bound*: for every deterministic H with $p_k \geq B_-$ for all $k \in H$, and every $a \in \{0, 1\}$,

$$\int \left(\widehat{\theta}_a(H) - \theta_a(H) \right)^2 dP^{\otimes n} \leq \frac{8 \sum_{k \in H} p_k}{m_1 \epsilon} + \frac{6}{m_1} + 4 \left(\frac{|H|}{u_n^2 B_-} \right)^2, \qquad u_n = \frac{m_1 - 2}{2} \epsilon.$$

Splitting about $\bar{\theta}_a(H)$ with $(x + y)^2 \leq 2x^2 + 2y^2$,

$$\left(\widehat{\theta}_a(H) - \theta_a(H) \right)^2 \leq 2 \left(\widehat{\theta}_a(H) - \bar{\theta}_a(H) \right)^2 + 2 \left(\bar{\theta}_a(H) - \theta_a(H) \right)^2.$$

(4a) The empirical-mass term. Put $g_{a,H}(O) = \mu_{aX} \mathbf{1}\{X \in H\}$, a function of one observation with values in $[0, 1]$. Grouping the observations of I_1 by category,

$$\bar{\theta}_a(H) = \frac{1}{m_1} \sum_{i \in I_1} g_{a,H}(O_i), \qquad \theta_a(H) = \mathbb{E}_P \left[g_{a,H}(O) \right],$$

so $\bar{\theta}_a(H) - \theta_a(H)$ is the centered sample mean of an iid bounded score over the m_1 observations of I_1, whence

$$\int \left(\bar{\theta}_a(H) - \theta_a(H) \right)^2 dP^{\otimes n} \leq \frac{\mathbb{E}_P \left[g_{a,H}(O)^2 \right]}{m_1} \leq \frac{1}{m_1},$$

the second inequality because $0 \leq g_{a,H} \leq 1$.

(4b) Exact decomposition of the ratio noise. Set

$$\eta_{ak}(O) = \mathbf{1}\{X = k,\ A = a\} \left(Y - \mu_{ak} \right), \qquad w_{ak} = \frac{N_k^{(1)}}{N_{ak}^{(1)}} \mathbf{1}\{N_{ak}^{(1)} > 0\},$$

and

$$R_a(H) = \sum_{k\in H} w_{ak} \sum_{i\in I_1} \eta_{ak}(O_i), \qquad \Delta_a(H) = \sum_{k\in H} \mu_{ak} N_k^{(1)} \mathbf{1}\{N_{ak}^{(1)} = 0\}.$$

Since $\sum_{i\in I_1} \eta_{ak}(O_i) = N_{ak1}^{(1)} - \mu_{ak} N_{ak}^{(1)}$, a term-by-term comparison of the two cases $N_{ak}^{(1)} > 0$ and $N_{ak}^{(1)} = 0$ gives the pointwise identity

$$\widehat{\theta}_a(H) - \bar{\theta}_a(H) = \frac{R_a(H) - \Delta_a(H)}{m_1}, \qquad \text{hence} \qquad \left(\widehat{\theta}_a(H) - \bar{\theta}_a(H)\right)^2 \le \frac{2R_a(H)^2}{m_1^2} + \frac{2\Delta_a(H)^2}{m_1^2}.$$

(4c) The residual term. All cross terms in the expansion of $R_a(H)^2$ integrate to zero: the weights w_{ak} depend on the sample only through the labels $(X_i, A_i)_{i\in I_1}$, and conditionally on those labels the residuals $\eta_{ak}(O_i)$ are independent across i with mean zero, while for $k \ne l$ the product $\eta_{ak}(O_i)\eta_{al}(O_i)$ vanishes identically because one observation lies in one category. Only the diagonal survives, and $|Y - \mu_{ak}| \le 1$ gives $\eta_{ak}(O_i)^2 \le \mathbf{1}\{X_i = k, A_i = a\}$, so that

$$\sum_{i\in I_1} w_{ak}^2\, \eta_{ak}(O_i)^2 \le w_{ak}^2 N_{ak}^{(1)} = \frac{\left(N_k^{(1)}\right)^2}{N_{ak}^{(1)}} \mathbf{1}\{N_{ak}^{(1)} > 0\}.$$

The nested-count second-moment identity for a multinomial category and its nested arm, evaluated with the category probability p_k and arm probability π_{ak} and then bounded using $\pi_{ak} \ge \epsilon$ from assumption 2, yields

$$\int \frac{\left(N_k^{(1)}\right)^2}{N_{ak}^{(1)}} \mathbf{1}\{N_{ak}^{(1)} > 0\}\, dP^{\otimes n} \le \frac{2m_1 p_k}{\pi_{ak}} \le \frac{2m_1 p_k}{\epsilon},$$

which requires $p_k > 0$ and is available here since $p_k \ge B_- > 0$ on H. Summing over $k \in H$,

$$\int R_a(H)^2\, dP^{\otimes n} \le \frac{2m_1}{\epsilon} \sum_{k\in H} p_k.$$

(4d) The missing-arm term. Since $0 \le \mu_{ak} \le 1$,

$$|\Delta_a(H)| \le \mathcal{N}_a(H) := \sum_{k\in H} N_k^{(1)} \mathbf{1}\{N_{ak}^{(1)} = 0\}, \qquad \text{so} \qquad \Delta_a(H)^2 \le \mathcal{N}_a(H)^2.$$

The second moment of $\mathcal{N}_a(H)$ is an exact multinomial expression in the masses p_k and arm probabilities π_{ak}: its diagonal part is $m_1 p_k (1-\pi_{ak})(1-p_k\pi_{ak})^{m_1-1} + m_1(m_1-1)\{p_k(1-\pi_{ak})\}^2 (1-p_k\pi_{ak})^{m_1-2}$ and its $k \ne l$ part is $m_1(m_1-1)p_k(1-\pi_{ak})p_l(1-\pi_{al})(1-p_k\pi_{ak}-p_l\pi_{al})^{m_1-2}$. Bounding $1-\pi_{ak} \le 1$, $m_1(m_1-1) \le m_1^2$, and $(1-p_k\pi_{ak})^{m_1-2} \le e^{-(m_1-2)\epsilon p_k}$ by $\pi_{ak} \ge \epsilon$, and writing $f_k = p_k e^{-u_n p_k}$ with $u_n = (m_1-2)\epsilon/2$, the diagonal is at most $m_1 p_k + m_1^2 f_k^2$ and each cross term at most $m_1^2 f_k f_l$, so that

$$\int \mathcal{N}_a(H)^2\, dP^{\otimes n} \le m_1 \sum_{k\in H} p_k + m_1^2 \left(\sum_{k\in H} f_k\right)^2 \le m_1 + m_1^2 \left(\sum_{k\in H} f_k\right)^2,$$

using $\sum_{k\in H} p_k \le 1$. Finally $s^2 \le e^s$ for $s \ge 0$, applied with $s = u_n p_k$, gives $f_k = p_k e^{-u_n p_k} \le 1/(u_n^2 p_k) \le 1/(u_n^2 B_-)$ for each $k \in H$, whence

$$\sum_{k\in H} f_k \le \frac{|H|}{u_n^2 B_-}, \qquad \text{and so} \qquad \int \Delta_a(H)^2 \, dP^{\otimes n} \le m_1 + m_1^2 \left(\frac{|H|}{u_n^2 B_-} \right)^2 .$$

(4e) Assembling the fixed-set arm bound. Integrating the pointwise bound of (4b) and inserting (4c) and (4d),

$$\begin{aligned} \int \left(\widehat{\theta}_a(H) - \bar{\theta}_a(H) \right)^2 dP^{\otimes n} &\le \frac{2}{m_1^2} \cdot \frac{2m_1}{\epsilon} \sum_{k\in H} p_k + \frac{2}{m_1^2} \left\{ m_1 + m_1^2 \left(\frac{|H|}{u_n^2 B_-} \right)^2 \right\} \\ &= \frac{4 \sum_{k\in H} p_k}{m_1 \epsilon} + \frac{2}{m_1} + 2 \left(\frac{|H|}{u_n^2 B_-} \right)^2 . \end{aligned}$$

Doubling this, adding twice the bound $1/m_1$ of (4a), and collecting terms gives exactly the fixed-set arm bound. (Note $m_1 \ge 4 > 2$, so $u_n > 0$.)

Step 5: the fixed-set envelope is at the target rate. Write

$$V = \kappa_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\} .$$

We claim that the right-hand side of the fixed-set arm bound is at most V for every H. Indeed:

$$\frac{8 \sum_{k\in H} p_k}{m_1 \epsilon} \le \frac{8}{m_1 \epsilon} \le \frac{16}{\epsilon n}, \qquad \frac{6}{m_1} \le \frac{12}{n},$$

using $\sum_{k\in H} p_k \le 1$ and $m_1 \ge n/2$. For the third term, $2m_0 \le n$ gives

$$B_- = \frac{256L}{2m_0} \ge \frac{256L}{n}, \qquad u_n = \frac{m_1 - 2}{2} \epsilon \ge \frac{n\epsilon}{8} \quad (\text{from } 4(m_1 - 2) \ge n),$$

hence $u_n^2 B_- \ge (n\epsilon/8)^2 \cdot 256L/n = 4n\epsilon^2 L$. With $|H| \le d$ and $0 < \log n \le L$,

$$4 \left(\frac{|H|}{u_n^2 B_-} \right)^2 \le 4 \left(\frac{d}{4n\epsilon^2 L} \right)^2 = \frac{1}{4\epsilon^4} \cdot \frac{d^2}{n^2 L^2} \le \frac{1}{\epsilon^4} \cdot \frac{d^2}{n^2 (\log n)^2} .$$

Adding the three bounds,

$$\frac{8 \sum_{k\in H} p_k}{m_1 \epsilon} + \frac{6}{m_1} + 4 \left(\frac{|H|}{u_n^2 B_-} \right)^2 \le \left(\frac{16}{\epsilon} + 12 \right) \frac{1}{n} + \frac{1}{\epsilon^4} \cdot \frac{d^2}{n^2 (\log n)^2} \le \kappa_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\} = V,$$

the last step because both $16/\epsilon + 12$ and $1/\epsilon^4$ are at most κ_ϵ and both rate terms are nonnegative. In particular $V \ge 0$.

Step 6: transfer to the random heavy set. Fix an arm a and let $\mathfrak{E} = \{H \subseteq \{1, \ldots, d\} : p_k \ge B_- \text{ for all } k \in H\}$. By Step 2, off $\mathcal{B}$ the realized set $\widehat{\mathcal{H}}_n$ belongs to $\mathfrak{E}$, so pointwise

$$\mathbf{1}_{\mathcal{B}^c} \left(\widehat{\theta}_a(\widehat{\mathcal{H}}_n) - \theta_a(\widehat{\mathcal{H}}_n) \right)^2 \le \sum_{H \in \mathfrak{E}} \mathbf{1}\{\widehat{\mathcal{H}}_n = H\} \left(\widehat{\theta}_a(H) - \theta_a(H) \right)^2 .$$

For each fixed H, the event $\{\widehat{\mathcal{H}}_n = H\}$ is a function of the pilot observations $(O_i)_{i\in I_0}$ alone, while $\left(\widehat{\theta}_a(H) - \theta_a(H)\right)^2$ is a function of the estimation observations $(O_i)_{i\in I_1}$ alone; under $P^{\otimes n}$ these two disjoint index blocks are independent, so the fiber integral factorizes,

$$\int \mathbf{1}\{\widehat{\mathcal{H}}_n = H\} \left(\widehat{\theta}_a(H) - \theta_a(H)\right)^2 dP^{\otimes n} = \mathrm{Pr}_{P^{\otimes n}}\left(\widehat{\mathcal{H}}_n = H\right) \int \left(\widehat{\theta}_a(H) - \theta_a(H)\right)^2 dP^{\otimes n}.$$

Every $H \in \mathfrak{E}$ satisfies the eligibility hypothesis of Step 4, so by Steps 4 and 5 the second factor is at most V. Summing over $H \in \mathfrak{E}$ and using $\sum_{H\in\mathfrak{E}} \mathrm{Pr}(\widehat{\mathcal{H}}_n = H) \le 1$ together with $V \ge 0$,

$$\int_{\mathcal{B}^c} \left(\widehat{\theta}_a(\widehat{\mathcal{H}}_n) - \theta_a(\widehat{\mathcal{H}}_n)\right)^2 dP^{\otimes n} \le V \qquad (a \in \{0,1\}).$$

Step 7: the pilot-good integral. Applying the pointwise bound of Step 3 with the random set $H = \widehat{\mathcal{H}}_n$ — legitimate because that bound holds for every realized set — and integrating over $\mathcal{B}^c$,

$$\int_{\mathcal{B}^c} \mathcal{D}^2 \, dP^{\otimes n} \le 2 \int_{\mathcal{B}^c} \left(\widehat{\theta}_1(\widehat{\mathcal{H}}_n) - \theta_1(\widehat{\mathcal{H}}_n)\right)^2 dP^{\otimes n} + 2 \int_{\mathcal{B}^c} \left(\widehat{\theta}_0(\widehat{\mathcal{H}}_n) - \theta_0(\widehat{\mathcal{H}}_n)\right)^2 dP^{\otimes n} \le 4V.$$

Step 8: conclusion. Since $n \ge 1$,

$$n^{-4} \le n^{-1} \le \frac{1}{n} + \frac{d^2}{n^2(\log n)^2},$$

so combining Step 1 with Step 7 in the decomposition of Step 0,

$$\int \mathcal{D}^2 \, dP^{\otimes n} \le C_{\mathrm{bad}} n^{-4} + 4V \le (C_{\mathrm{bad}} + 4\kappa_\epsilon) \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} = C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\},$$

which, since $\mu_n = P^{\otimes n}$, is the asserted bound. □

lemma 3 matches the light-cell rate from lemma 1. Combining the two controls the untruncated sum $\widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}}$; the final truncation in definition 11 cannot increase squared error against an ATE lying in the binary-outcome range.

The lower bound is transferred from the one-sided subclass in which only treated conditional means vary. This subclass is contained in the observational ATE experiment class and identifies the same hard functional through $\tau(P)$. The external minimax input is the fixed-sample large-alphabet lower bound of Zeng et al. [2024]; standard lower-bound comparisons then pass it to definition 14, in the same spirit as classical minimax reductions [Tsybakov, 2009, Cai and Low, 2011].

Lemma 4. [`lem:ate-lower-bound-transfer`] (ATE Minimax Lower Bound)[2] *For any* $0 < \epsilon < 1/2$, *let* $\mathsf{R}_{n,d,\epsilon}$ *denote the minimax mean-squared-error risk of definition 14. There exist constants* $a_\epsilon, b_\epsilon \in \mathbb{R}$ *and a natural-number cutoff* N_ϵ *such that* $a_\epsilon > 0$, $b_\epsilon > 0$, *and for every* $n, d \in \mathbb{N}$ *with* $d > 0$, $n \ge N_\epsilon$, *and*

$$d \le b_\epsilon n \log n,$$

one has

$$\mathsf{R}_{n,d,\epsilon} \ge a_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Proof of lemma 4. Fix $0 < \epsilon < 1/2$, and write $\mathcal{O}_n := (\{1,\ldots,d\} \times \{0,1\} \times \{0,1\})^n$ for the finite sample space, so that a measurable estimator is any map $\widehat{\tau} : \mathcal{O}_n \to \mathbb{R}$ and E_P denotes expectation under $P^{\otimes n}$. A supremum over an empty index set is read as 0, the convention already in force in definition 14.

Step 1: the control-zero subclass and its treated-arm risk. Define the subclass of laws whose control regression vanishes in every category,

$$\mathcal{D}_{n,d,\epsilon} := \left\{P : P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon},\ \mu_{0k} = 0 \text{ for every } k = 1,\ldots,d\right\},$$

the treated-arm functional

$$\psi_1(P) := \sum_{k=1}^{d} p_k \mu_{1k},$$

and the treated-arm minimax risk over that subclass,

$$\mathsf{R}^{(1)}_{n,d,\epsilon} := \inf_{\widehat{\tau}} \sup_{P \in \mathcal{D}_{n,d,\epsilon}} E_P\left[(\widehat{\tau} - \psi_1(P))^2\right],$$

the infimum again ranging over all measurable estimators based on $O_1,\ldots,O_n$, exactly as in definition 14. The cited fixed-sample control-zero lower bound [Zeng et al., 2024] is imported in the following form: there exist constants $a_\epsilon > 0$, $b_\epsilon > 0$ and a cutoff $N_\epsilon \in \mathbb{N}$, depending only on ϵ, such that

$$a_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \le \mathsf{R}^{(1)}_{n,d,\epsilon} \qquad \text{whenever } d > 0,\ n \ge N_\epsilon,\ d \le b_\epsilon n \log n.$$

Step 2: a uniform risk envelope. For every law P satisfying assumption 2 and every estimator $\widehat{\tau}$,

$$E_P\left[(\widehat{\tau} - \tau(P))^2\right] \le (S(\widehat{\tau}) + 1)^2, \qquad S(\widehat{\tau}) := \sum_{o \in \mathcal{O}_n} |\widehat{\tau}(o)|.$$

Indeed, the weighted-regression form of definition 6 gives $\tau(P) = \sum_{k=1}^{d} p_k(\mu_{1k} - \mu_{0k})$ with $\sum_{k=1}^{d} p_k = 1$, $p_k \ge 0$ and $|\mu_{1k} - \mu_{0k}| \le 1$, whence

$$|\tau(P)| \le 1;$$

and $|\widehat{\tau}(o)| \le S(\widehat{\tau})$ for each $o \in \mathcal{O}_n$, since $S(\widehat{\tau})$ is a finite sum of nonnegative terms one of which is $|\widehat{\tau}(o)|$. Therefore $|\widehat{\tau}(o) - \tau(P)| \le S(\widehat{\tau}) + 1$ pointwise on $\mathcal{O}_n$, and integrating the squared bound against the probability law $P^{\otimes n}$ gives the display. Two consequences are used below: for each fixed $\widehat{\tau}$, both the supremum over $\mathcal{E}_{n,d,\epsilon}$ and the supremum over $\mathcal{D}_{n,d,\epsilon}$ are taken over sets of reals bounded above by $(S(\widehat{\tau}) + 1)^2$, so both are genuine finite suprema.

Step 3: the target identity on the subclass. Let $P \in \mathcal{D}_{n,d,\epsilon}$. Then P satisfies assumption 2, so the weighted-regression form of definition 6 applies, and $\mu_{0k} = 0$ for every k gives

$$\tau(P) = \sum_{k=1}^{d} p_k(\mu_{1k} - \mu_{0k}) = \sum_{k=1}^{d} p_k \mu_{1k} = \psi_1(P).$$

[2]**Formalization scope.** This result uses the published conclusion [Zeng et al., 2024] (Theorem 2; Appendix C.5; Lemma 4; Appendix D.2). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

Step 4: the risk comparison. We claim

$$\mathsf{R}^{(1)}_{n,d,\epsilon} \le \mathsf{R}_{n,d,\epsilon}.$$

Fix a measurable estimator $\widehat{\tau}$ and split on whether the subclass is inhabited.

If no law belongs to $\mathcal{D}_{n,d,\epsilon}$, then $\sup_{P\in\mathcal{D}_{n,d,\epsilon}} E_P[(\widehat{\tau}-\psi_1(P))^2] = 0$ by the empty-supremum convention, and the right-hand worst-case risk is nonnegative: if $\mathcal{E}_{n,d,\epsilon}$ is likewise empty it too equals 0, and otherwise it dominates the single nonnegative value $E_P[(\widehat{\tau}-\tau(P))^2] \ge 0$ attained at any member $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, that domination being legitimate because Step 2 makes the supremum finite.

If instead $\mathcal{D}_{n,d,\epsilon}$ contains a law, take any $P \in \mathcal{D}_{n,d,\epsilon}$. By definition of $\mathcal{D}_{n,d,\epsilon}$ we have $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, and by Step 3 the treated-arm loss of $\widehat{\tau}$ at P is its ATE loss, so

$$E_P\left[(\widehat{\tau}-\psi_1(P))^2\right] = E_P\left[(\widehat{\tau}-\tau(P))^2\right] \le \sup_{Q^{\otimes n}\in\mathcal{E}_{n,d,\epsilon}} E_Q\left[(\widehat{\tau}-\tau(Q))^2\right],$$

the last step again using the finiteness supplied by Step 2. Taking the supremum over $P \in \mathcal{D}_{n,d,\epsilon}$,

$$\sup_{P\in\mathcal{D}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}-\psi_1(P))^2\right] \le \sup_{P^{\otimes n}\in\mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}-\tau(P))^2\right].$$

Both cases give this inequality for every measurable $\widehat{\tau}$. Moreover the left-hand side is bounded below by 0 uniformly in $\widehat{\tau}$: it is 0 when the subclass is empty, and otherwise it dominates the nonnegative squared-error risk at any one member of the subclass, whose supremum is finite by Step 2. Hence the infima over measurable estimators may be compared termwise, which is the claimed inequality $\mathsf{R}^{(1)}_{n,d,\epsilon} \le \mathsf{R}_{n,d,\epsilon}$.

Step 5: conclusion. Let $a_\epsilon, b_\epsilon, N_\epsilon$ be the constants of Step 1 and let $n, d \in \mathbb{N}$ satisfy $d > 0$, $n \ge N_\epsilon$ and $d \le b_\epsilon n \log n$. Chaining the cited bound of Step 1 with the comparison of Step 4,

$$a_\epsilon\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \le \mathsf{R}^{(1)}_{n,d,\epsilon} \le \mathsf{R}_{n,d,\epsilon},$$

which is the asserted lower bound, with the same constants $a_\epsilon > 0$, $b_\epsilon > 0$ and cutoff N_ϵ. □

lemma 4 is the only auxiliary statement in this section whose substantive lower-bound scale is imported from the published large-alphabet ATE construction [Zeng et al., 2024]. Its conclusion is stated directly for the minimax risk in definition 14, so it can be paired with the hybrid upper bound without changing the experiment class.

The remaining statements describe the randomized endpoint and the envelope near it. At exact randomization, treatment assignment carries no category-specific imbalance, and the centered linear estimator in definition 12 becomes the natural comparison estimator.

Lemma 5. [`lem:near-randomization-linear-upper`] (Centered Linear MSE Bound) *For any $n, d \in \mathbb{N}$, any $\epsilon \in \mathbb{R}$, any single-observation law P, and any sample law μ_n, suppose that*

- ***(Sample size.)*** $n > 0$.
- ***(Experiment class.)*** *μ_n belongs to the experiment class $\mathcal{E}_{n,d,\epsilon}$ generated by P, as in definition 3.*

Define

$$\widehat{\tau}_{\mathrm{ctr}} = \frac{1}{n}\sum_{i=1}^{n} 2(2A_i - 1)\left(Y_i - \frac{1}{2}\right).$$

Then, with $\tau(P)$ as in definition 6,

$$\mathbb{E}_{\mu_n}\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right] \le \frac{1}{n} + 4\left(\frac{1}{2} - \epsilon\right)^2.$$

Proof of lemma 5. Membership of μ_n in $\mathcal{E}_{n,d,\epsilon}$ supplies $\mu_n = P^{\otimes n}$, the range restriction $0 < \epsilon \le 1/2$, and the overlap bounds of assumption 2 for P; these are used throughout.

Write

$$S(O) = 2(2A-1)\left(Y - \frac{1}{2}\right), \qquad \widehat{\tau}_{\mathrm{ctr}} = \frac{1}{n}\sum_{i=1}^{n} S(O_i).$$

Since A and Y are binary, $S(O) = 1$ when $A = Y$ and $S(O) = -1$ when $A \ne Y$; in particular $S(O)^2 = 1$ pointwise, so

$$\mathrm{Var}_P(S) = E_P\left[S(O)^2\right] - \{E_P S(O)\}^2 = 1 - \{E_P S(O)\}^2 \le 1.$$

Under $\mu_n = P^{\otimes n}$ the observations $O_1, \ldots, O_n$ are iid, so the variance of their average is the one-observation variance divided by n:

$$\mathrm{Var}_{\mu_n}(\widehat{\tau}_{\mathrm{ctr}}) = \frac{\mathrm{Var}_P(S)}{n} \le \frac{1}{n}.$$

It remains to bound the bias. Evaluating S on the four atoms of a category, its value is $+1$ at $(a, y) \in \{(1,1), (0,0)\}$ and -1 at $(a, y) \in \{(1,0), (0,1)\}$, so summing the atom masses against these values,

$$E_P S(O) = \sum_{k=1}^{d} (q_{1k1} - q_{1k0} + q_{0k0} - q_{0k1}).$$

Fix k with $p_k > 0$. Overlap together with $\epsilon > 0$ gives $0 < \epsilon \le \pi_k \le 1 - \epsilon < 1$, so both arm masses $q_{1k0} + q_{1k1} = \pi_k p_k$ and $q_{0k0} + q_{0k1} = (1 - \pi_k)p_k$ are strictly positive and μ_{1k}, μ_{0k} are the genuine conditional means of Y in the two arms. Consequently

$$q_{1k1} = \mu_{1k}\pi_k p_k, \quad q_{1k0} = (1 - \mu_{1k})\pi_k p_k, \quad q_{0k1} = \mu_{0k}(1 - \pi_k)p_k, \quad q_{0k0} = (1 - \mu_{0k})(1 - \pi_k)p_k,$$

and substituting these four expressions and rearranging yields the exact conditional-bias identity

$$q_{1k1} - q_{1k0} + q_{0k0} - q_{0k1} = p_k(\mu_{1k} - \mu_{0k}) + 2\left(\pi_k - \tfrac{1}{2}\right)p_k(\mu_{1k} + \mu_{0k} - 1).$$

If instead $p_k = 0$, then the four nonnegative masses $q_{0k0}, q_{0k1}, q_{1k0}, q_{1k1}$ sum to 0 and therefore all vanish, so the left-hand side above is 0; the right-hand side is 0 as well, both of its terms carrying the factor $p_k = 0$. The displayed identity therefore holds for every k.

Summing the identity over k and subtracting $\tau(P) = \sum_{k=1}^{d} p_k(\mu_{1k} - \mu_{0k})$ from definition 6, which is the form the ATE functional takes under assumption 2, gives

$$E_P S(O) - \tau(P) = \sum_{k=1}^{d} 2\left(\pi_k - \tfrac{1}{2}\right)p_k(\mu_{1k} + \mu_{0k} - 1).$$

For a category with $p_k > 0$, the overlap bounds $\epsilon \le \pi_k \le 1-\epsilon$ give $|\pi_k - \frac{1}{2}| \le \frac{1}{2} - \epsilon$, while $\mu_{1k}, \mu_{0k} \in [0,1]$ gives $|\mu_{1k} + \mu_{0k} - 1| \le 1$; since $\frac{1}{2} - \epsilon \ge 0$ and $p_k \ge 0$, multiplying these bounds yields

$$\left|2\left(\pi_k - \tfrac{1}{2}\right) p_k(\mu_{1k} + \mu_{0k} - 1)\right| \le 2\left(\frac{1}{2} - \epsilon\right) p_k,$$

and for $p_k = 0$ both sides vanish, so the bound holds for every k. The triangle inequality applied to the previous display, followed by $\sum_{k=1}^{d} p_k = 1$ (the categories partition the sample space), gives

$$|E_P S(O) - \tau(P)| \le \sum_{k=1}^{d} 2\left(\frac{1}{2} - \epsilon\right) p_k = 2\left(\frac{1}{2} - \epsilon\right).$$

Finally, $n > 0$ and $\mu_n = P^{\otimes n}$ give

$$E_{\mu_n} \widehat{\tau}_{\text{ctr}} = \frac{1}{n} \sum_{i=1}^{n} E_P S(O_i) = E_P S(O),$$

so the bias of $\widehat{\tau}_{\text{ctr}}$ is exactly the quantity just bounded, and since $2(\frac{1}{2} - \epsilon) \ge 0$ the bound squares to

$$\{E_{\mu_n} \widehat{\tau}_{\text{ctr}} - \tau(P)\}^2 \le 4\left(\frac{1}{2} - \epsilon\right)^2.$$

The mean-squared error splits as

$$E_{\mu_n}\left[(\widehat{\tau}_{\text{ctr}} - \tau(P))^2\right] = \mathrm{Var}_{\mu_n}(\widehat{\tau}_{\text{ctr}}) + \{E_{\mu_n} \widehat{\tau}_{\text{ctr}} - \tau(P)\}^2,$$

and adding the variance bound to the squared-bias bound gives

$$E_{\mu_n}\left[(\widehat{\tau}_{\text{ctr}} - \tau(P))^2\right] \le \frac{1}{n} + 4\left(\frac{1}{2} - \epsilon\right)^2.$$

□

lemma 5 quantifies the cost of using the centered estimator away from exact randomization. The variance term is parametric, while the bias term is controlled by the distance from $\epsilon = 1/2$. This bound complements the fixed-interior hybrid analysis in overlap classes close to randomized treatment.

A matching parametric lower bound at the endpoint follows already from a one-category binary-outcome subproblem.

Lemma 6. [`lem:one-category-bernoulli-lower`] (Bernoulli Randomization Lower Bound) *For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, the minimax mean-squared-error risk $\mathsf{R}_{n,d,1/2}$ of definition 14 satisfies*

$$\mathsf{R}_{n,d,1/2} \ge \frac{1}{100n}.$$

Proof of lemma 6. Fix $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, and put

$$\delta = \frac{2/5}{\sqrt{n}}.$$

Since $\sqrt{n} \geq 1$,

$$0 \leq \delta \leq \frac{2}{5}, \qquad \delta^2 = \frac{4}{25n}, \qquad \frac{1}{2} + \delta \leq \frac{9}{10} \leq 1.$$

Step 1: the two competing laws. Define two laws $P_0, P_1 \in \mathsf{Prob}(\{1, \ldots, d\} \times \{0,1\}^2)$ as in definition 1 by prescribing, for every category $k \in \{1, \ldots, d\}$, the four-cell vector $q_k = (q_{0k0}, q_{0k1}, q_{1k0}, q_{1k1})$ of definition 1, whose entries are the cell masses q_{aky} of $(X, A, Y) = (k, a, y)$ listed in the treatment–outcome order $(a, y) = (0,0), (0,1), (1,0), (1,1)$:

$$q_k(P_0) = \left(\frac{1}{4d}, \frac{1}{4d}, \frac{1}{4d}, \frac{1}{4d}\right), \qquad q_k(P_1) = \left(\frac{1}{4d}, \frac{1}{4d}, \frac{1}{2d}\left(\frac{1}{2} - \delta\right), \frac{1}{2d}\left(\frac{1}{2} + \delta\right)\right).$$

Both prescriptions are legitimate laws: all coordinates are nonnegative because $0 \leq \delta \leq 2/5 < 1/2$, and in each case the $4d$ masses sum to $d \cdot \{1/(4d) + 1/(4d) + 1/(2d)\} = 1$. Equivalently, under both laws X is uniform on $\{1, \ldots, d\}$ and $A \mid X = k \sim \mathsf{Bernoulli}(1/2)$ in the sense of definition 2; the control arm has $Y \mid A = 0, X = k \sim \mathsf{Bernoulli}(1/2)$ under both laws, while the treated arm has $Y \mid A = 1, X = k \sim \mathsf{Bernoulli}(1/2)$ under P_0 and $Y \mid A = 1, X = k \sim \mathsf{Bernoulli}(1/2 + \delta)$ under P_1.

Step 2: both product laws lie in the class. For $j \in \{0,1\}$ and every k, summing the four displayed coordinates and the two treated coordinates gives

$$p_k = \sum_{a,y \in \{0,1\}} q_{aky}(P_j) = \frac{1}{d} > 0, \qquad \pi_k = \frac{q_{1k0}(P_j) + q_{1k1}(P_j)}{p_k} = \frac{1/(2d)}{1/d} = \frac{1}{2}.$$

Hence $1/2 \leq \pi_k \leq 1 - 1/2$ for every category, so each P_j satisfies assumption 2 with overlap constant $\epsilon = 1/2$, and therefore

$$P_0^{\otimes n} \in \mathcal{E}_{n,d,1/2}, \qquad P_1^{\otimes n} \in \mathcal{E}_{n,d,1/2}$$

in the sense of definition 3.

Step 3: the two estimand values. The conditional outcome means of definition 1 are read off the same coordinates: for every k,

$$\mu_{0k}(P_j) = \frac{q_{0k1}(P_j)}{q_{0k0}(P_j) + q_{0k1}(P_j)} = \frac{1}{2} \quad (j = 0,1), \qquad \mu_{1k}(P_0) = \frac{1/(4d)}{1/(2d)} = \frac{1}{2}, \qquad \mu_{1k}(P_1) = \frac{(1/(2d))(1/2 + \delta)}{1/(2d)} = \frac{1}{2}$$

Substituting into the weighted-regression form of definition 6 and using $p_k = 1/d$,

$$\tau(P_0) = \sum_{k=1}^{d} p_k \{\mu_{1k}(P_0) - \mu_{0k}(P_0)\} = 0, \qquad \tau(P_1) = \sum_{k=1}^{d} \frac{1}{d} \delta = \delta.$$

Step 4: the two sample laws are statistically indistinguishable. For probability laws μ, ν on a common finite alphabet $\mathcal{Z}$ with ν charging every point, write

$$\chi^2(\mu \| \nu) = \sum_{z \in \mathcal{Z}} \frac{\{\mu(z) - \nu(z)\}^2}{\nu(z)}, \qquad \mathrm{TV}(\mu, \nu) = \sup_{A \subseteq \mathcal{Z}} |\mu(A) - \nu(A)|.$$

The laws P_0 and P_1 differ only in the two treated cells of each category, where the masses differ by $\pm\delta/(2d)$, and P_0 puts mass $1/(4d)$ on each of those cells; summing over the $2d$ treated cells gives the single-observation value

$$\chi^2(P_1 \| P_0) = 2d \cdot \frac{\{\delta/(2d)\}^2}{1/(4d)} = 2\delta^2 = \frac{(1/2)\,\delta^2}{(1/2)(1 - 1/2)}.$$

Because the n observations are independent and identically distributed under each product law, the shifted divergence tensorizes, and $1 + x \le e^x$ gives

$$1 + \chi^2\big(P_1^{\otimes n} \,\|\, P_0^{\otimes n}\big) = \big\{1 + \chi^2(P_1 \| P_0)\big\}^n = \big(1 + 2\delta^2\big)^n \le \exp\big(2n\delta^2\big)\,.$$

With $\delta^2 = 4/(25n)$ the exponent is calibrated below $\log 2$:

$$2n\delta^2 = \frac{8}{25} \le \log 2,$$

since $\log 2 > 0.693 > 8/25$. Hence $1 + \chi^2(P_1^{\otimes n} \| P_0^{\otimes n}) \le 2$, that is,

$$\chi^2\big(P_1^{\otimes n} \,\|\, P_0^{\otimes n}\big) \le 1,$$

and the Cauchy–Schwarz comparison $\mathrm{TV}(\mu, \nu) \le \frac{1}{2}\sqrt{\chi^2(\mu\|\nu)}$ yields

$$\mathrm{TV}\big(P_0^{\otimes n}, P_1^{\otimes n}\big) \le \frac{1}{2}\sqrt{1} = \frac{1}{2}.$$

Step 5: two-point testing bound. Let $\widehat{\tau}$ be any measurable estimator based on $O_1, \ldots, O_n$, as in definition 14. By Step 3 the two estimand values are separated at radius $\delta/2$:

$$2 \cdot \frac{\delta}{2} = \delta = |\tau(P_0) - \tau(P_1)|\,.$$

No number is within $\delta/2$ of both $\tau(P_0)$ and $\tau(P_1)$, so the two events $\{|\widehat{\tau} - \tau(P_0)| < \delta/2\}$ and $\{|\widehat{\tau} - \tau(P_1)| < \delta/2\}$ are disjoint; comparing the two laws on these events and using the bound of Step 4 gives

$$\max_{j=0,1} P_j^{\otimes n}\left(|\widehat{\tau} - \tau(P_j)| \ge \frac{\delta}{2}\right) \ge \frac{1 - \mathrm{TV}\big(P_0^{\otimes n}, P_1^{\otimes n}\big)}{2} \ge \frac{1 - 1/2}{2} = \frac{1}{4}.$$

Step 6: from miss probability to mean squared error. For each $j = 0, 1$, the miss event coincides with $\big\{(\widehat{\tau} - \tau(P_j))^2 \ge (\delta/2)^2\big\}$, and the squared loss is nonnegative, so integrating it over that event gives

$$\left(\frac{\delta}{2}\right)^2 P_j^{\otimes n}\left(|\widehat{\tau} - \tau(P_j)| \ge \frac{\delta}{2}\right) \le E_{P_j^{\otimes n}}\big[(\widehat{\tau} - \tau(P_j))^2\big]\,.$$

Multiplying the bound of Step 5 by $(\delta/2)^2 \ge 0$ and using $\delta^2 = 4/(25n)$,

$$\max_{j=0,1} E_{P_j^{\otimes n}}\big[(\widehat{\tau} - \tau(P_j))^2\big] \ge \left(\frac{\delta}{2}\right)^2 \cdot \frac{1}{4} = \frac{\delta^2}{16} = \frac{1}{100n}.$$

Step 7: passing to the minimax risk. By Step 2 both $P_0^{\otimes n}$ and $P_1^{\otimes n}$ are members of $\mathcal{E}_{n,d,1/2}$, and for a fixed estimator the risks over the class are bounded above uniformly, because the sample space is finite and $|\tau(P)| \le 1$ for every law in the class. Hence the worst-case risk of $\widehat{\tau}$ dominates each of the two risks in Step 6:

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,1/2}} E_{P^{\otimes n}}\big[(\widehat{\tau} - \tau(P))^2\big] \ge \max_{j=0,1} E_{P_j^{\otimes n}}\big[(\widehat{\tau} - \tau(P_j))^2\big] \ge \frac{1}{100n}.$$

The estimator $\widehat{\tau}$ was an arbitrary measurable estimator, so taking the infimum over all such estimators in definition 14 gives

$$\mathsf{R}_{n,d,1/2} \geq \frac{1}{100n},$$

which is the claim.

□

The one-category reduction in lemma 6 removes the large-alphabet feature entirely, leaving the ordinary difficulty of estimating a Bernoulli mean contrast under randomized assignment. Combining it with the centered-estimator upper bound gives the endpoint bracket.

Lemma 7. [`lem:randomized-endpoint-minimax`] (Randomized Endpoint Minimax Risk) *For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, the minimax mean-squared-error risk of definition 14 at exact randomization satisfies*

$$\frac{1}{100n} \leq \mathsf{R}_{n,d,1/2} \leq \frac{1}{n}.$$

Proof of lemma 7. Fix $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$. The two halves of the bracket are proved separately.

Lower bound. The hypotheses $n > 0$ and $d > 0$ fixed here are exactly those of lemma 6, which therefore gives

$$\frac{1}{100n} \leq \mathsf{R}_{n,d,1/2}.$$

Upper bound. For a measurable estimator $\widehat{\tau}$ based on $O_1, \dots, O_n$, write

$$\mathsf{W}(\widehat{\tau}) := \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,1/2}} E_{P^{\otimes n}}\left[(\widehat{\tau} - \tau(P))^2\right]$$

for its worst-case risk over the experiment class of definition 3 at $\epsilon = 1/2$, with $\tau(P)$ the ATE functional of definition 6; by definition 14, $\mathsf{R}_{n,d,1/2} = \inf_{\widehat{\tau}} \mathsf{W}(\widehat{\tau})$, the infimum ranging over all measurable estimators.

Step 1: the infimum may be compared with a single candidate. For every measurable $\widehat{\tau}$ and every $P^{\otimes n} \in \mathcal{E}_{n,d,1/2}$, the risk $E_{P^{\otimes n}}[(\widehat{\tau} - \tau(P))^2] = \int (\widehat{\tau} - \tau(P))^2 \, dP^{\otimes n}$ is the integral of a nonnegative function and hence nonnegative, so

$$0 \leq \mathsf{W}(\widehat{\tau}) \qquad \text{for every measurable } \widehat{\tau},$$

with the supremum read as 0 when $\mathcal{E}_{n,d,1/2}$ is empty. The set of worst-case risks appearing in definition 14 is therefore bounded below, so its infimum is no larger than the value at any one measurable candidate.

Step 2: the candidate. Take the centered estimator of definition 12,

$$\widehat{\tau}_{\mathrm{ctr}} = \frac{1}{n} \sum_{i=1}^{n} 2(2A_i - 1) \left(Y_i - \frac{1}{2} \right),$$

which is a fixed real-valued function of the sample $O_1, \dots, O_n$ alone; since each O_i takes finitely many values, this function is measurable, so $\widehat{\tau}_{\mathrm{ctr}}$ is an admissible candidate. By Step 1,

$$\mathsf{R}_{n,d,1/2} \leq \mathsf{W}(\widehat{\tau}_{\mathrm{ctr}}).$$

Step 3: the worst-case risk of the candidate. If $\mathcal{E}_{n,d,1/2}$ is empty, then $\mathsf{W}(\widehat{\tau}_{\mathrm{ctr}}) = 0 \le 1/n$. Otherwise, let $P^{\otimes n} \in \mathcal{E}_{n,d,1/2}$ be arbitrary, generated by the single-observation law P. Both hypotheses of lemma 5 hold for the sample law $\mu_n = P^{\otimes n}$, namely $n > 0$ and membership of μ_n in the class generated by P, so with overlap constant $\epsilon = 1/2$ it gives

$$E_{P^{\otimes n}}\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right] \le \frac{1}{n} + 4\left(\frac{1}{2} - \frac{1}{2}\right)^2 = \frac{1}{n}.$$

The right-hand side does not depend on P, so taking the supremum over the class,

$$\mathsf{W}(\widehat{\tau}_{\mathrm{ctr}}) \le \frac{1}{n}.$$

Chaining Step 2 with Step 3 gives $\mathsf{R}_{n,d,1/2} \le 1/n$, and together with the lower bound,

$$\frac{1}{100n} \le \mathsf{R}_{n,d,1/2} \le \frac{1}{n}.$$

□

lemma 7 shows that the endpoint $\epsilon = 1/2$ is qualitatively different from the fixed-interior large-alphabet problem: uniformly in d, the risk remains of order n^{-1}. The next lemma combines this observation with the hybrid upper bound through the oracle-calibrated selected estimator introduced in definition 13.

Lemma 8. [`lem:combined-upper-envelope`] (Combined Upper Envelope) *Fix constants $\epsilon, C_\epsilon, \rho_\epsilon \in \mathbb{R}$ and $N_\epsilon \in \mathbb{N}$. Assume:*

- ***(Interior overlap.)*** $0 < \epsilon < 1/2$.
- ***(Positive constants.)*** $C_\epsilon > 0$ *and* $\rho_\epsilon > 0$.
- ***(Hybrid bound.)*** *For every $n, d \in \mathbb{N}$, if $0 < n$, $N_\epsilon \le n$, and $d \le \rho_\epsilon n \log n$, then the hybrid estimator $\widehat{\tau}_n^{\mathrm{hyb}}$ of definition 11 satisfies*

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(P)\right)^2\right] \le C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Then for every $n, d \in \mathbb{N}$ with $0 < n$, $N_\epsilon \le n$, and $d \le \rho_\epsilon n \log n$, let $K_\epsilon = \max\{C_\epsilon, 4\}$. Define the C_ϵ, ϵ-dependent selected estimator $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$ to be $\widehat{\tau}_n^{\mathrm{hyb}}$ when

$$C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \le \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and to be $\widehat{\tau}_{\mathrm{ctr}}$ otherwise. This selected estimator satisfies

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P)\right)^2\right] \le K_\epsilon \left[\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2\right\}\right].$$

Proof of lemma 8. Fix $n, d \in \mathbb{N}$ with $0 < n$, $N_\epsilon \le n$ and $d \le \rho_\epsilon n \log n$, and abbreviate

$$a = \frac{1}{n}, \qquad u = \frac{d^2}{n^2(\log n)^2}, \qquad v = \left(\frac{1}{2} - \epsilon\right)^2, \qquad K_\epsilon = \max\{C_\epsilon, 4\},$$

so that the hybrid rate is $a + u$ and the asserted envelope is $K_\epsilon\{a + \min(u, v)\}$. Because $n > 0$, we have $a > 0$; $u \ge 0$ as a ratio of squares; and $v \ge 0$ as a square. Also $C_\epsilon \le K_\epsilon$ and $4 \le K_\epsilon$ by the definition of the maximum.

Step 1: the centered estimator. With $\widehat{\tau}_{\text{ctr}}$ as in definition 12,

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[(\widehat{\tau}_{\text{ctr}} - \tau(P))^2\right] \le a + 4v.$$

If $\mathcal{E}_{n,d,\epsilon}$ is empty the supremum is 0, and $0 \le a + 4v$ because $a > 0$ and $v \ge 0$. Otherwise, for each $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, lemma 5, applied with $\mu_n = P^{\otimes n}$ and the overlap constant ϵ, gives

$$\mathbb{E}_{P^{\otimes n}}\left[(\widehat{\tau}_{\text{ctr}} - \tau(P))^2\right] \le \frac{1}{n} + 4\left(\frac{1}{2} - \epsilon\right)^2 = a + 4v,$$

and taking the supremum over the class preserves the bound.

Step 2: the selected estimator. We claim

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}^{\text{sel}}_{C_\epsilon,\epsilon} - \tau(P)\right)^2\right] \le \min\left\{C_\epsilon(a+u),\, a + 4v\right\}.$$

By definition 13 the selector splits on the comparison $C_\epsilon(a + u) \le a + 4v$, so we treat the two cases separately.

If $C_\epsilon(a + u) \le a + 4v$, then $\widehat{\tau}^{\text{sel}}_{C_\epsilon,\epsilon} = \widehat{\tau}^{\text{hyb}}_n$, and the same comparison identifies the minimum,

$$\min\left\{C_\epsilon(a+u),\, a + 4v\right\} = C_\epsilon(a + u).$$

Since $0 < n$, $N_\epsilon \le n$ and $d \le \rho_\epsilon n \log n$, the assumed hybrid bound applies and yields

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}^{\text{hyb}}_n - \tau(P)\right)^2\right] \le C_\epsilon\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} = C_\epsilon(a + u),$$

which is the claim in this case.

If instead $C_\epsilon(a+u) > a+4v$, then $\widehat{\tau}^{\text{sel}}_{C_\epsilon,\epsilon} = \widehat{\tau}_{\text{ctr}}$, and the reversed comparison $a+4v \le C_\epsilon(a+u)$ identifies the minimum,

$$\min\left\{C_\epsilon(a+u),\, a + 4v\right\} = a + 4v,$$

so the claim is exactly the bound of Step 1.

Step 3: comparison with the minimax risk. The sample $O_1, \ldots, O_n$ takes values in the finite set $(\{1, \ldots, d\} \times \{0, 1\}^2)^n$, so every real-valued function of it, in particular $\widehat{\tau}^{\text{sel}}_{C_\epsilon,\epsilon}$, is measurable and is therefore one of the candidates over which the infimum in definition 14 is taken. That infimum is over a set of reals bounded below by 0: for a fixed measurable $\widehat{\tau}$, each $\mathbb{E}_{P^{\otimes n}}[(\widehat{\tau} - \tau(P))^2]$ is nonnegative as the integral of a square, and the supremum over an empty class is 0. Hence

$$\mathsf{R}_{n,d,\epsilon} \le \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}^{\text{sel}}_{C_\epsilon,\epsilon} - \tau(P)\right)^2\right].$$

Step 4: the minimum is below the envelope. It remains to show

$$\min\{C_\epsilon(a+u),\, a+4v\} \le K_\epsilon\{a+\min(u,v)\}.$$

Suppose first $u \le v$, so $\min(u,v) = u$. Since $a+u \ge 0$ and $C_\epsilon \le K_\epsilon$,

$$\min\{C_\epsilon(a+u),\, a+4v\} \le C_\epsilon(a+u) \le K_\epsilon(a+u) = K_\epsilon\{a+\min(u,v)\}.$$

Suppose instead $v < u$, so $\min(u,v) = v$. From $4 \le K_\epsilon$ we get both $a \le K_\epsilon a$ (as $1 \le 4 \le K_\epsilon$ and $a \ge 0$) and $4v \le K_\epsilon v$ (as $v \ge 0$); adding the two and factoring gives

$$a + 4v \le K_\epsilon a + K_\epsilon v = K_\epsilon(a+v),$$

whence

$$\min\{C_\epsilon(a+u),\, a+4v\} \le a+4v \le K_\epsilon(a+v) = K_\epsilon\{a+\min(u,v)\}.$$

Chaining Step 2 with Step 4 and substituting back the definitions of a, u, v gives

$$\sup_{P^{\otimes n}\in\mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P)\right)^2\right] \le K_\epsilon\left[\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, \left(\frac{1}{2}-\epsilon\right)^2\right\}\right],$$

which is the asserted bound; combined with Step 3 it also bounds $\mathsf{R}_{n,d,\epsilon}$ by the same envelope. □

lemma 8 formalizes the oracle-calibrated deterministic upper-envelope comparison between the fixed-interior hybrid estimator and the centered near-randomization estimator. Consequently, theorem 2 gives a theoretical risk guarantee near $\epsilon = 1/2$. Data-driven adaptation and the sharp triangular-array minimax transition are identified as future directions in the limitations section.

B verification note

This appendix documents the scope and boundary of the machine-checked component of the paper.

The checked development includes the finite-alphabet observed-data model used in assumption 1 and definition 3. In particular, the single-observation law P, the iid product law $P^{\otimes n}$, and the finite four-cell representation by category are represented directly. The same layer also includes the bridge from finite products to the infinite-product notation used for sequences of observations, so that the sample-level risk statements are connected to repeated iid sampling rather than to an informal limiting construction.

The local observational model is checked at the level needed for the overlap and ATE objects. The overlap cone in definition 4, the homogeneous cell functional in definition 5, and the ATE functional in definition 6 are represented as finite-alphabet objects with arbitrary category masses. Thus the verification covers the unrestricted discrete-confounding geometry used throughout the paper, including categories with small marginal probability, subject only to the stated overlap restrictions.

The estimator objects are also included in the checked scope. This covers the deterministic split, the split cell counts, the heavy and light pilot sets, the ratio contribution on pilot-certified heavy categories, the Chebyshev-polynomial continuation on light categories, the factorial-moment estimator, and the final truncation in definition 11. The minimax risk in definition 14 is checked as a worst-case mean-squared-error criterion over the experiment class, with the infimum taken over measurable estimators as stated there.

Lean declaration names and low-level convention checks are retained in the formal crosswalk and verification artifacts. The statistical proof prose focuses on the heavy/light decomposition, approximation bias, variance control, minimax reduction, and endpoint comparison.

For results whose theorem-local verification footnotes cite an external mathematical input, the cited input is part of the paper's stated dependency boundary. The machine check covers the formal statement that imports that input, the compatibility of its hypotheses with the objects defined in the paper, and the remaining derivation from that point. Results without such a footnote are checked internally against the definitions and assumptions stated in the paper.

All theorem statements and formal derivations in this paper are machine-checked in Lean 4, subject to the theorem-local external inputs explicitly identified in the formalization-scope footnotes.

C Proofs of the main results

Proof of theorem 1. Fix $0 < \epsilon < 1/2$. Throughout, $n \log n \geq 0$ for every sample size, so an inequality of the form $d \leq \rho\, n \log n$ is preserved when ρ is replaced by any larger constant.

1. **The upper half.** We first prove the following claim. There are constants $C_U, \rho_U > 0$ and an integer $N_U < \infty$, depending only on ϵ, such that for every n, d, every single-observation law P with $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, every $n \geq N_U$ and every

$$d \leq \rho_U n \log n,$$

one has

$$\mathsf{R}_{n,d,\epsilon} \leq \sup_{Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_Q\left[\left(\widehat{\tau}_n^{\text{hyb}} - \tau(Q)\right)^2\right] \leq C_U \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Constants. Since $0 < \epsilon \leq 1/2$, lemma 1 applies; write $C_L, c_L > 0$ for the two constants it supplies. Since $0 < \epsilon < 1/2$, lemma 3 applies; write $C_H, \rho_H > 0$ and $N_H < \infty$ for the constants and cutoff it supplies. Set

$$C_U = 2(C_L + C_H), \qquad \rho_U = \min\{c_L, \rho_H\}, \qquad N_U = N_H.$$

Then $C_U > 0$ and $\rho_U > 0$, and the hypothesis $d \leq \rho_U n \log n$ yields both growth gates,

$$d \leq \rho_U n \log n \leq c_L n \log n, \qquad d \leq \rho_U n \log n \leq \rho_H n \log n.$$

The truncation cannot increase squared error. For every real x and every $t \in [-1, 1]$,

$$(\max\{-1, \min(1, x)\} - t)^2 \leq (x - t)^2.$$

Indeed, if $x \leq -1$ the left-hand entry is -1, which lies between x and t; if $x > 1$ it is 1, which again lies between x and t; otherwise it is x and the two sides agree. Moreover, by definition 6 the estimand is $\tau(P) = \sum_{k=1}^{d} p_k(\mu_{1k} - \mu_{0k})$ with $\sum_{k=1}^{d} p_k = 1$ and $\mu_{0k}, \mu_{1k} \in [0, 1]$, whence

$$\tau(P) \in [-1, 1].$$

Applying the previous display with $x = \widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}}$ and $t = \tau(P)$, and using the definition of $\widehat{\tau}_n^{\text{hyb}}$ in definition 11, gives the sample-by-sample bound

$$\left(\widehat{\tau}_n^{\text{hyb}} - \tau(P)\right)^2 \leq \left(\widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}} - \tau(P)\right)^2.$$

Splitting the target. The pilot-light set is the complement of the pilot-heavy set, $\widehat{\mathcal{L}}_n = \widehat{\mathcal{H}}_n^c$, so the two index sets are disjoint with union $\{1, \dots, d\}$ and definition 6 gives, sample by sample,

$$\sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k) + \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) = \sum_{k=1}^{d} \phi(q_k) = \tau(P).$$

Substituting this identity and using $(u+v)^2 \leq 2u^2 + 2v^2$, which is $0 \leq (u-v)^2$, we obtain sample by sample

$$\begin{aligned}\left(\widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}} - \tau(P)\right)^2 &= \left\{\left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k)\right) + \left(\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)\right)\right\}^2 \\ &\leq 2\left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k)\right)^2 + 2\left(\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)\right)^2.\end{aligned}$$

Integration and the two component bounds. The observation alphabet is finite, so every function of the sample appearing above is bounded and integrable, and integrating the previous two displays against $P^{\otimes n}$ preserves the inequality and splits the right-hand side by linearity:

$$E_P\left[\left(\widehat{\tau}_n^{\text{hyb}} - \tau(P)\right)^2\right] \leq 2E_P\left[\left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k)\right)^2\right] + 2E_P\left[\left(\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)\right)^2\right].$$

Nothing in this derivation, nor in the two growth gates displayed above, refers to the particular law: the gates constrain only n and d, and the constants $C_L, c_L, C_H, \rho_H, N_H$ depend only on ϵ. Hence for *every* $Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ the heavy bound of lemma 3 (available because $n \geq N_U = N_H$ and $d \leq \rho_H n \log n$) and the light bound of lemma 1 (available because $d \leq c_L n \log n$) may be inserted, giving

$$\begin{aligned}E_Q\left[\left(\widehat{\tau}_n^{\text{hyb}} - \tau(Q)\right)^2\right] &\leq 2C_H\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} + 2C_L\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \\ &= C_U\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.\end{aligned}$$

The experiment class is nonempty, since $P^{\otimes n}$ belongs to it, so taking the supremum over $Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ gives the second inequality of the claim.

The minimax risk does not exceed the hybrid worst case. The sample space $(\{1, \dots, d\} \times \{0,1\}^2)^n$ is finite, so $\widehat{\tau}_n^{\text{hyb}}$ is a measurable estimator and is therefore one of the competitors in definition 14. For any measurable $\widehat{\tau}$ and any $Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, the bound $\tau(Q) \in [-1, 1]$

obtained above gives $|\widehat{\tau}(x)-\tau(Q)| \le \sum_{x'} |\widehat{\tau}(x')|+1$ at every sample point x, where x' ranges over the finite sample space, and hence

$$E_Q\left[(\widehat{\tau}-\tau(Q))^2\right] \le \left(\sum_{x'} |\widehat{\tau}(x')|+1\right)^2,$$

so each estimator's worst-case risk is a finite nonnegative number and the infimum in definition 14 is taken over a nonempty set of reals bounded below by 0. Evaluating that infimum at $\widehat{\tau}_n^{\mathrm{hyb}}$ gives the first inequality of the claim, and the claim is proved.

2. **The lower half and the constants of the theorem.** Lemma 4, which transfers the fixed-sample one-arm lower bound of [Zeng et al., 2024, Theorem 2; Appendix C.5; Lemma 4; Appendix D.2], supplies constants $a>0$ and $\rho_L>0$ and an integer cutoff $N_L<\infty$ such that its displayed lower bound for $\mathsf{R}_{n,d,\epsilon}$ holds whenever $d>0$, $n\ge N_L$ and $d\le \rho_L n\log n$. Now define

$$a_\epsilon = a, \qquad \rho_\epsilon=\min\{\rho_U,\rho_L\}, \qquad C_\epsilon = C_U, \qquad N_\epsilon=\max\{N_U,N_L\}.$$

These constants are positive, and $N_\epsilon<\infty$.

3. **Sharp sandwich.** Assume $d>0$, $n\ge N_\epsilon$ and $d\le\rho_\epsilon n\log n$. Then $n\ge N_U$ and $n\ge N_L$, while

$$d\le \rho_\epsilon n\log n\le \rho_U n\log n, \qquad d\le \rho_\epsilon n\log n\le \rho_L n\log n,$$

so both growth gates are in force. Step 2 and Step 1 therefore give

$$a_\epsilon\left\{\frac{1}{n}+\frac{d^2}{n^2(\log n)^2}\right\} \le \mathsf{R}_{n,d,\epsilon} \le \sup_{Q^{\otimes n}\in\mathcal{E}_{n,d,\epsilon}} E_Q\left[\left(\widehat{\tau}_n^{\mathrm{hyb}}-\tau(Q)\right)^2\right] \le C_\epsilon\left\{\frac{1}{n}+\frac{d^2}{n^2(\log n)^2}\right\},$$

which is the sharp sandwich.

4. **Parametric window.** Fix $K>0$ and put

$$C_K = C_U(1+K^2)>0.$$

Assume $d>0$, $n\ge N_\epsilon$, $d\le K\sqrt{n}\log n$ and $d\le \rho_\epsilon n\log n$.

Because $K\sqrt{n}\ge 0$ and $d>0$, the inequality $d\le K\sqrt{n}\log n$ forces $\log n>0$; in particular $n\ge 2$, so n and $n^2(\log n)^2$ are strictly positive. Both sides of $d\le K\sqrt{n}\log n$ are nonnegative, so squaring is legitimate and $(\sqrt{n})^2=n$ gives

$$d^2\le K^2 n(\log n)^2.$$

Dividing by $n^2(\log n)^2>0$,

$$\frac{d^2}{n^2(\log n)^2}\le\frac{K^2}{n}.$$

For the lower bound, $d^2/\{n^2(\log n)^2\}\ge 0$ and the retained gate $d\le\rho_\epsilon n\log n\le\rho_L n\log n$ with $d>0$ and $n\ge N_L$ let us invoke Step 2:

$$\frac{a_\epsilon}{n}\le a_\epsilon\left\{\frac{1}{n}+\frac{d^2}{n^2(\log n)^2}\right\}\le \mathsf{R}_{n,d,\epsilon}.$$

For the upper bound, the same gate gives $d \le \rho_U n \log n$ and $n \ge N_U$, so Step 1 applies; combining it with the displayed ratio bound,

$$\mathsf{R}_{n,d,\epsilon} \le C_U \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \le C_U \left(\frac{1}{n} + \frac{K^2}{n} \right) = \frac{C_U(1+K^2)}{n} = \frac{C_K}{n}.$$

This proves the parametric window.

5. **Consistency threshold.** Let n_j, d_j satisfy $n_j \to \infty$, $d_j > 0$ eventually, and $d_j \le \rho_\epsilon n_j \log n_j$ eventually. Set

$$q_j = \frac{d_j}{n_j \log n_j}.$$

For every j, squaring the definition of q_j gives the identity

$$\frac{1}{n_j} + \frac{d_j^2}{n_j^2(\log n_j)^2} = \frac{1}{n_j} + q_j^2.$$

Since $n_j \to \infty$ we have $1/n_j \to 0$; and eventually $n_j \ge 2$, so eventually $\log n_j > 0$ and therefore $q_j \ge 0$.

Consider those indices j — all sufficiently large ones — for which simultaneously $d_j > 0$, $d_j \le \rho_\epsilon n_j \log n_j$ and $n_j \ge N_\epsilon$. Step 1 requires a member of the experiment class to be exhibited, and $d_j > 0$ permits one: let $P_d^\star$ be the law on $\{1, \dots, d\} \times \{0,1\} \times \{0,1\}$ with

$$P_d^\star(X = k, A = a, Y = y) = \frac{1}{4d} \qquad \text{for all } k \in \{1, \dots, d\},\ a, y \in \{0,1\},$$

that is, $p_k = 1/d$, $\pi_k = 1/2$ and $\mu_{0k} = \mu_{1k} = 1/2$ in every category. Because $0 < \epsilon < 1/2$ we have $\epsilon \le 1/2 \le 1 - \epsilon$, so $\pi_k \in [\epsilon, 1-\epsilon]$ for every category, assumption 2 holds with overlap constant ϵ, and $(P_{d_j}^\star)^{\otimes n_j} \in \mathcal{E}_{n_j,d_j,\epsilon}$. Applying Step 2 and then Step 1 with this law, and rewriting by the identity above, gives for those same indices

$$a_\epsilon \left(\frac{1}{n_j} + q_j^2 \right) \le \mathsf{R}_{n_j,d_j,\epsilon} \le C_\epsilon \left(\frac{1}{n_j} + q_j^2 \right).$$

Suppose first that $\mathsf{R}_{n_j,d_j,\epsilon} \to 0$. Since $a_\epsilon > 0$, the left inequality above reads

$$0 \le \frac{1}{n_j} + q_j^2 \le \frac{1}{a_\epsilon} \mathsf{R}_{n_j,d_j,\epsilon}$$

eventually, and the right side tends to 0, so $1/n_j + q_j^2 \to 0$. Then $0 \le q_j^2 \le n_j^{-1} + q_j^2$ gives $q_j^2 \to 0$, and since $\sqrt{q_j^2} = |q_j| = q_j$ eventually, continuity of the square-root map at 0 gives $q_j \to 0$.

Conversely, suppose $q_j \to 0$. Then $q_j^2 \to 0$, and together with $1/n_j \to 0$ this yields

$$\frac{1}{n_j} + q_j^2 \to 0.$$

The minimax risk is nonnegative, because in definition 14 it is an infimum of worst-case mean squared errors, each of which is nonnegative; combined with the right inequality of the two-sided bound above this squeezes $\mathsf{R}_{n_j,d_j,\epsilon} \to 0$. This proves the consistency threshold and completes the proof.

$\square$

Proof of theorem 2. Throughout, for $n, d \in \mathbb{N}$ write

$$r_{n,d} = \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}$$

for the target rate, and, for a single-observation law P and a sample $O_1, \ldots, O_n$, write

$$\mathcal{D}_{\mathcal{H}} = \widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k), \qquad \mathcal{D}_{\mathcal{L}} = \widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)$$

for the errors of the two branches of definition 11 against their pilot-selected targets, with $\widehat{T}_{\mathcal{H}}$, $\widehat{T}_{\mathcal{L}}$, $\widehat{\mathcal{H}}_n$, $\widehat{\mathcal{L}}_n$ as defined there and ϕ as in definition 5. Both $\mathcal{D}_{\mathcal{H}}$ and $\mathcal{D}_{\mathcal{L}}$ are functions of the sample alone.

1. **(Computability.)** The first assertion of lemma 1 is precisely the computability clause: there is a universal constant $K > 0$ such that, for every n and d, a real-arithmetic program reading the hybrid count vector returns $\widehat{\tau}_n^{\mathrm{hyb}}$ exactly on every sample $O_1, \ldots, O_n$ and uses at most $K\, d\, M(n)^4$ arithmetic and comparison operations. It is proved there with the explicit value $K = 450$, which is a positive integer, so the clause holds with that K.

2. **(Calibration of the constants.)** Fix $0 < \epsilon < 1/2$. Lemma 1, applicable because $0 < \epsilon \leq 1/2$, supplies constants $C_L, c_L > 0$ depending only on ϵ such that $\mathbb{E}_{P^{\otimes n}}[\mathcal{D}_{\mathcal{L}}^2] \leq C_L\, r_{n,d}$ for every $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ with $d \leq c_L n \log n$. Lemma 3, applicable because $0 < \epsilon < 1/2$, supplies constants $C_H, c_H > 0$ and an integer $N_H < \infty$, again depending only on ϵ, such that $\mathbb{E}_{P^{\otimes n}}[\mathcal{D}_{\mathcal{H}}^2] \leq C_H\, r_{n,d}$ for every $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ with $n \geq N_H$ and $d \leq c_H n \log n$. Set

$$C_\epsilon = 2(C_L + C_H), \qquad \rho_\epsilon = \min\{c_L, c_H\}, \qquad N_\epsilon = N_H.$$

   Then $C_\epsilon > 0$ and $\rho_\epsilon > 0$. Fix $n, d \in \mathbb{N}$ with $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$. Since $n \geq 1$ gives $n \log n \geq 0$, and since $\rho_\epsilon \leq c_L$ and $\rho_\epsilon \leq c_H$, the growth hypothesis transfers to both component lemmas:

$$d \leq \rho_\epsilon n \log n \leq c_L n \log n, \qquad d \leq \rho_\epsilon n \log n \leq c_H n \log n.$$

3. **(Hybrid envelope.)** Let $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ be arbitrary, so that P satisfies assumption 2 with constant ϵ and the sample law is the product law of definition 3. We first bound the squared error pointwise in the sample.

   Because $\widehat{\mathcal{L}}_n$ is the complement of $\widehat{\mathcal{H}}_n$ in $\{1, \ldots, d\}$, the two pilot-selected targets add up to the estimand: by definition 6,

$$\sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k) + \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) = \sum_{k=1}^{d} \phi(q_k) = \tau(P),$$

   for every realization of the pilot split.

Next, the estimand lies in the range enforced by the truncation. By definition 6, $\tau(P) = \sum_k p_k(\mu_{1k} - \mu_{0k})$; the category masses satisfy $p_k \geq 0$ and $\sum_k p_k = 1$, and each conditional mean lies in $[0, 1]$, so $|\mu_{1k} - \mu_{0k}| \leq 1$. Hence

$$|\tau(P)| \leq \sum_{k=1}^{d} p_k|\mu_{1k} - \mu_{0k}| \leq \sum_{k=1}^{d} p_k = 1, \qquad \text{i.e.} \qquad \tau(P) \in [-1, 1].$$

Consequently the truncation in definition 11 cannot increase the squared error: for every $x \in \mathbb{R}$ and every $t \in [-1, 1]$,

$$(\max\{-1, \min(1, x)\} - t)^2 \leq (x - t)^2.$$

Indeed, if $-1 \leq x \leq 1$ the two sides are equal; if $x > 1$ the left-hand side is $(1 - t)^2$ and $0 \leq 1 - t \leq x - t$; and if $x < -1$ the left-hand side is $(-1 - t)^2$ with $x - t \leq -1 - t \leq 0$.

Applying this with $x = \widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}}$ and $t = \tau(P)$, so that $\max\{-1, \min(1, x)\} = \widehat{\tau}_n^{\text{hyb}}$, then substituting the target identity and using $(\alpha - \beta)^2 \geq 0$, we obtain pointwise in the sample

$$\left(\widehat{\tau}_n^{\text{hyb}} - \tau(P)\right)^2 \leq \left(\widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}} - \tau(P)\right)^2 = (\mathcal{D}_{\mathcal{H}} + \mathcal{D}_{\mathcal{L}})^2 \leq 2\,\mathcal{D}_{\mathcal{H}}^2 + 2\,\mathcal{D}_{\mathcal{L}}^2.$$

All the quantities involved are functions of a sample from a finite alphabet, hence bounded and integrable, so integrating this inequality against $P^{\otimes n}$ and using linearity gives

$$\mathbb{E}_{P^{\otimes n}}\left[(\widehat{\tau}_n^{\text{hyb}} - \tau(P))^2\right] \leq 2\,\mathbb{E}_{P^{\otimes n}}\left[\mathcal{D}_{\mathcal{H}}^2\right] + 2\,\mathbb{E}_{P^{\otimes n}}\left[\mathcal{D}_{\mathcal{L}}^2\right].$$

The two component bounds of Step 2 are in force at this n, d, and law, so the right-hand side is at most

$$2C_H\, r_{n,d} + 2C_L\, r_{n,d} = C_\epsilon\, r_{n,d}.$$

The constants C_L, C_H do not depend on the member of the class, so the same bound holds for every $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ and therefore for the supremum:

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[(\widehat{\tau}_n^{\text{hyb}} - \tau(P))^2\right] \leq C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

This is the first display of the fixed-overlap clause, and it holds for every $n \geq N_\epsilon$ and every d with $d \leq \rho_\epsilon n \log n$.

4. **(Selected estimator.)** Keep n, d as in Step 2. The selection rule of definition 13 compares the two real numbers $C_\epsilon\{1/n + d^2/(n^2 \log^2 n)\}$ and $1/n + 4(1/2 - \epsilon)^2$, neither of which depends on the data; $\widehat{\tau}_{C_\epsilon,\epsilon}^{\text{sel}}$ is therefore equal to $\widehat{\tau}_n^{\text{hyb}}$ or to $\widehat{\tau}_{\text{ctr}}$ as a fixed function of the sample, and in either case it is a measurable estimator based on $O_1, \ldots, O_n$. Since the infimum in definition 14 ranges over all such estimators, it is no larger than the worst-case risk of this one:

$$\mathsf{R}_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[(\widehat{\tau}_{C_\epsilon,\epsilon}^{\text{sel}} - \tau(P))^2\right].$$

For the envelope itself, the hypotheses of lemma 8 have all been verified: $0 < \epsilon < 1/2$ by assumption, $C_\epsilon > 0$ and $\rho_\epsilon > 0$ by Step 2, and the hybrid worst-case bound at every

$n, d \in \mathbb{N}$ with $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$ by Step 3. Its selected estimator is the one just described, formed from the same C_ϵ and ϵ, and it yields

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2 \right] \leq \max\{C_\epsilon, 4\} \left\{ \frac{1}{n} + \min\left[\frac{d^2}{n^2(\log n)^2}, (1/2-\epsilon)^2 \right] \right\},$$

which is the remaining display of the fixed-overlap clause with $K_\epsilon = \max\{C_\epsilon, 4\}$.

5. **(Centered estimator.)** Now fix $0 < \epsilon \leq 1/2$ and $n, d \in \mathbb{N}$. If $\mathcal{E}_{n,d,\epsilon}$ has no members, its supremum is 0 and the asserted bound holds because $1/n + 4(1/2-\epsilon)^2 \geq 0$. Otherwise, let $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$. The hypotheses of lemma 5 hold, since $n > 0$ and $P^{\otimes n}$ is a member of the experiment class generated by P, so

$$\mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2 \right] \leq \frac{1}{n} + 4\left(\frac{1}{2} - \epsilon \right)^2,$$

with $\widehat{\tau}_{\mathrm{ctr}}$ the estimator of definition 12. The right-hand side does not depend on the member, so taking the supremum over the class gives the centered-estimator clause.

6. **(Endpoint bracket.)** Finally, for every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, lemma 7 gives the endpoint clause directly:
$$\frac{1}{100n} \leq \mathsf{R}_{n,d,1/2} \leq \frac{1}{n}.$$
Together with Steps 1, 3, 4, and 5 this proves the theorem.

□

References

Sivaraman Balakrishnan, Edward H. Kennedy, and Larry Wasserman. The fundamental limits of structure-agnostic functional estimation. *arXiv preprint arXiv:2305.04116*, 2023. doi: 10.48550/arXiv.2305.04116. To appear in Statistical Science.

Heejung Bang and James M. Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005. doi: 10.1111/j.1541-0420.2005.00377.x.

Alexandre Belloni, Victor Chernozhukov, and Christian Hansen. High-dimensional methods and inference on structural and treatment effects. *Journal of Economic Perspectives*, 28(2):29–50, 2014. doi: 10.1257/jep.28.2.29.

T. Tony Cai and Mark G. Low. Testing composite hypotheses, hermite polynomials and optimal estimation of a nonsmooth functional. *The Annals of Statistics*, 39(2):1012–1041, 2011. doi: 10.1214/10-AOS849.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097.

Alexander D'Amour, Peng Ding, Avi Feller, Lihua Lei, and Jasjeet Sekhon. Overlap in observational studies with high-dimensional covariates. *Journal of Econometrics*, 221(2):644–654, 2021. doi: 10.1016/j.jeconom.2019.10.014.

Jinyong Hahn. On the role of the propensity score in efficient semiparametric estimation of average treatment effects. *Econometrica*, 66(2):315–331, 1998. doi: 10.2307/2998560.

Yanjun Han, Jiantao Jiao, and Tsachy Weissman. Minimax estimation of discrete distributions under <inline-formula> <tex-math notation=”latex”>ℓ_1 </tex-math></inline-formula> loss. *IEEE Transactions on Information Theory*, 61(11): 6343–6354, 2015. doi: 10.1109/TIT.2015.2478816.

Yanjun Han, Jiantao Jiao, and Tsachy Weissman. Minimax estimation of divergences between discrete distributions. *IEEE Journal on Selected Areas in Information Theory*, 1(3):814–823, 2020. doi: 10.1109/JSAIT.2020.3041036.

Keisuke Hirano, Guido W. Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 71(4):1161–1189, 2003. doi: 10.1111/1468-0262.00442.

Jiantao Jiao, Kartik Venkat, Yanjun Han, and Tsachy Weissman. Minimax estimation of functionals of discrete distributions. *IEEE Transactions on Information Theory*, 61(5):2835–2885, 2015. doi: 10.1109/TIT.2015.2412945.

Jikai Jin and Vasilis Syrgkanis. Structure-agnostic optimality of doubly robust learning for treatment effect estimation. *arXiv preprint arXiv:2402.14264*, 2025. doi: 10.48550/arXiv.2402.14264. To appear in COLT 2025.

Edward H. Kennedy, Sivaraman Balakrishnan, James M. Robins, and Larry Wasserman. Minimax rates for heterogeneous causal effect estimation. *The Annals of Statistics*, 52(2):793–816, 2024. doi: 10.1214/24-AOS2369.

O. Lepski, A. Nemirovski, and V. Spokoiny. On estimation of the l r norm of a regression function. *Probability Theory and Related Fields*, 113(2):221–253, 1999. doi: 10.1007/s004409970006.

Lin Liu, Rajarshi Mukherjee, Whitney K. Newey, and James M. Robins. Semiparametric efficient empirical higher order influence function estimators. *arXiv preprint arXiv:1705.07577*, 2017. doi: 10.48550/arXiv.1705.07577.

Alon Orlitsky, Ananda Theertha Suresh, and Yihong Wu. Optimal prediction of the number of unseen species. *Proceedings of the National Academy of Sciences*, 113(47):13283–13288, 2016. doi: 10.1073/pnas.1607774113.

Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6): 1191–1253, 2003. doi: 10.1162/089976603321780272.

James M. Robins and Ya'acov Ritov. Toward a curse of dimensionality appropriate (CODA) asymptotic theory for semi-parametric models. *Statistics in Medicine*, 16(3):285–319, 1997. doi: 10.1002/(SICI)1097-0258(19970215)16:3⟨285::AID-SIM535⟩3.0.CO;2-\#.

James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994. doi: 10.1080/01621459.1994.10476818.

James M. Robins, Lingling Li, Eric J. Tchetgen Tchetgen, and Aad W. van der Vaart. Higher order influence functions and minimax estimation of nonlinear functionals. In *Probability and Statistics: Essays in Honor of David A. Freedman*, volume 2 of *Institute of Mathematical Statistics Collections*, pages 335–421. Institute of Mathematical Statistics, Beachwood, OH, 2008. doi: 10.1214/193940307000000527.

James M. Robins, Eric J. Tchetgen Tchetgen, Lingling Li, and Aad W. van der Vaart. Semiparametric minimax rates. *Electronic Journal of Statistics*, 3:1305–1321, 2009. doi: 10.1214/09-EJS479.

James M. W. Robins, Lingling Li, Rajarshi Mukherjee, Eric J. Tchetgen Tchetgen, and Aad van der Vaart. Minimax estimation of a functional on a structured high-dimensional model. *The Annals of Statistics*, 45(5):1951–1987, 2017. doi: 10.1214/16-AOS1515.

Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41.

Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974. doi: 10.1037/h0037350.

Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009. doi: 10.1007/b13794.

Gregory Valiant and Paul Valiant. Estimating the unseen. In *Proceedings of the 43rd Annual ACM Symposium on Theory of Computing*, pages 685–694, 2011. doi: 10.1145/1993636.1993727.

Gregory Valiant and Paul Valiant. Estimating the unseen. *Journal of the ACM*, 64(6):37:1–37:41, 2017. doi: 10.1145/3125643.

Mark J. van der Laan and Daniel Rubin. Targeted maximum likelihood learning. *The International Journal of Biostatistics*, 2(1), 2006. doi: 10.2202/1557-4679.1043.

Yihong Wu and Pengkun Yang. Minimax rates of entropy estimation on large alphabets via best polynomial approximation. *IEEE Transactions on Information Theory*, 62(6):3702–3720, 2016. doi: 10.1109/TIT.2016.2548468.

Yihong Wu and Pengkun Yang. Chebyshev polynomials, moment matching, and optimal estimation of the unseen. *The Annals of Statistics*, 47(2):857–883, 2019. doi: 10.1214/17-AOS1665.

Zhenghao Zeng, Sivaraman Balakrishnan, Yanjun Han, and Edward H. Kennedy. Causal inference with high-dimensional discrete covariates. *arXiv preprint arXiv:2405.00118*, 2024. doi: 10.48550/arXiv.2405.00118. Version 3.